MAGNETOM

Safety-Related Tests Procedure System

Safety-Related Tests according to IEC 62353

Avanto
Espree
The protocol M6-020.834.05.02.XX is required for these instructions

Document Version

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1.1 Safety Information

WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

1.2 General remarks

Based on the risk analysis and the clinical evaluation, the manufacturer is obligated to determine the type and frequency of repair activities as well as the inspection measures required for products to ensure continuously safe and proper operation.

The safety-related tests

- are recurring tests. The scope and schedule of these tests are defined by the manufacturer.
- should determine whether the affected product is in good operating condition, ensuring the safety of patients, users, and other individuals.
- do not replace maintenance activities.

Country-specific laws and regulations have to be taken into consideration, e.g. in Germany the “Medizinprodukte-Betreiberverordung (MPBetreibV)” (Medical Device Operator Regulations), IEC 62353 / DIN EN 62353, etc.

These safety-related tests correspond to the technical safety checks according to “MPBetreibV” (Medical Device Operator Regulations) “Sicherheitstechnische Kontrolle” including IEC 62353 / DIN EN 62353.

The safety-related checks have to be performed **during installation** and at the **intervals** defined in the time schedule.

1.3 Documentation

A protocol of the safety-related tests has to be generated. All fields have to be completed. If a field does not apply to the system or if there is no information to be entered, enter 'n.a.' in the field.

Each test performed successfully has to be confirmed in the test protocol by placing a check mark next to "O.K." Tests that cannot be performed (e.g. options) or which are not performed (e.g. interval is more than one year) should be marked with "n.a." If an error occurs which cannot be eliminated, mark the respective test with "not O.K."

The protocol of the installation "First measurement" serves as a reference document for subsequent repeat measurements. For this reason, it has to be archived by the operator in the System Owner Manual.

The MR quality department needs the protocol for legal reasons. Send a copy of the original to:

Tab. 1 Address MR Quality Department

Siemens
Healthcare
MR Quality Department
P.O. BOX 3260
91050 Erlangen
Germany
mail to: immrqpinstallations.healthcare@siemens.com

The protocols of repeat measurements should be filed in the System Owner Manual as well until subsequent safety-related tests have been performed.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products e.g. in the event damages occur.

1.4 Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

1.5 Time Schedule for the Safety-Related Tests

Tab. 2 General tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
System	12	20	
Options and accessories	12	20	
Phantoms	12	10	
Quench tube	12	40	Quench tube outlet sign if missing, sign is part of the sign set 1-7 Adhesive tape for quench tube if missing, material number. 3866493
Magnet	12	50	Helium leak detector
Patient fan	12	10	Air filter
User documentation	12	5	
Icon and button labels	12	5	
Controlled access area	12	10	Sign if missing, sign is part of the sign set 1-7
RF room door	12	5	
Warning signs	12	9	
Annual average time including all options: 184 min (3.1 h)			

Tab. 3 Electrical tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Measurement of the system ground wire	24	30	Ground test meter
Measurement of the MRSC ground wire	12	10	Ground test meter
Annual average time including all options: 25 min (0.5 h)			

Tab. 4 Mechanical tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
TALES	24	70	
Annual average time including all options: 35 min (0.6 h)			

Tab. 5 Function tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
EPO (if installed)	12	15	
Gradient supervision	12	40	Smoke wick p.n. 10684293 or Test spray p.n. 10684091
Patient table	12	40	
Patient Trolley	12	10	
MSUP type K2256: ERDU Test	12	30	ERDU test load MSUP-Display Unit
MSUP type K2306: Magnet Stop button test	12	15	
RF Integrity	12	5	
QA	12	70 - 120[1]	QA Phantom set Service plug[2]
Annual average time including all options[3]: 230 min (4.0h)			

1. Depending on installed options (calculation with 90 min)
2. Avanto only if the multi nucleus spectroscopy option MNO is installed
3. Depending on installed options

2.1 System

Required time: 20 min

Required tools: n.a.

SI Visual inspection of system

- Perform the general visual inspection of the system.
  - » The system may not contain damages that impair safety.
  - » There must be no undefined parts inside the magnet room. They could be magnetic. Call the user to clarify before any movement of those parts.
- Correct visible defects with respect to safety prior performing further service works.

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

2.2 Options and accessories

Required time: 20 min

Required tools:n.a.

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator / user, determine whether additional options / accessories (e.g., a trolley from an old system) are in use.

  » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the protocol.

2.2.1 Patient trolley (optional)

- Thoroughly inspect the trolley frame, the wheels, interlocking mechanism, the brake system, and the hydraulic system for damage, and take corrective action, if necessary.

In order to document which trolley has been inspected, enter the material and the serial number of the inspected options in the protocol.

2.2.2 Surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the protocol.

2.2.3 List of options and accessories

> i This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, Trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

SI Head coil

SI Neck coil

SI Body matrix coil

SI Spine matrix coil

SI Extremity Coil

SI Breast coil

SI TX / RX head coil

SI Loop coil

SI Flex coil interface

SI Flex coil small

SI Flex coil large

SI Wrist coil

SI Foot ankle coil

SI PA matrix coil

SI Knee coil

2.2.3.1 Option MRSC

SI Option MRSC

2.2.3.2 Option Trolley

SI Option Trolley

2.2.3.3 Option 1

> If an option (for example an another coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

SI Option 1

2.2.3.4 Option 2

SI Option 2

2.2.3.5 Option 3

SI Option 3

2.2.3.6 Option 4

SI Option 4

2.2.3.7 Option 5

SI Option 5

2.3 Phantoms

Required time: 10 min

Required tools: n.a.

SI Checking the phantoms

Please read the “MR-Specific Safety Information” prior to inspecting the phantoms.

1. Inspect every phantom used on-site for cracks, leaks, or extensive wear and tear.
   » Replace the phantom if there is any evidence of damage that may affect normal operation.

Defective phantoms must be disposed of. Refer to the “Disposal Instructions”.

2. Inspect every phantom used on-site for excessive air bubbles. Air bubbles can influence your Tune-up results and quality measurements.
3. Ensure that fluid is at an adequate level so that measurement performance is not adversely affected.

Water mixture phantoms with filler cap:

- Refill the phantom, if necessary.
- Phantoms with distilled water mixture - fill with distilled water only.

Water mixture phantoms without filler cap (welded):

- Refilling is not possible. Water mixture phantoms without filler cap must be replaced

Oil phantoms without filler cap (welded):

- Refilling is not possible. Oil phantoms without filler cap must be replaced

Oil phantoms with filler cap:

- Refilling is not possible. Oil phantoms with filler cap must be replaced.

  Do not open the screw cap. It will lead to leakage.

  For example spherical phantom blue D240 p/n 7577666 or p/n 10496685 : The diameter of the air bubble should not exceed 12 cm.

2.4 Quench tube

2.4.1 Quench tube inspection

Required time: 40 min

Required tools: If missing from the quench tube the following spare parts are available from the SPC and must be fitted:-

- Adhesive tape for Quench tube p.n. 3866493
- Quench outlet warning sign.

SI Visual inspection of the quench tube

- Visually inspect the condition of the quench tube.

2.4.2 Internal quench tube inspection

Check the following (visible inspection from the ground only, it is not necessary to remove ceiling tiles etc):

- Check that no unauthorized changes to the design of the quench tube **have been made**, both inside or outside the RF room.
- Check the **quench tube** and **insulation** for **damage**.
- Check that the **connection screws** to the magnet quench valve are all fitted and tight.
- Check the warning **label or tape** is attached to the **tube insulation.**
  - **On long tube sections** the warning labels or tape should be at regular distances of **between 2 - 3 meters**.

Fig. 1: Warning label example- Vent tube

Fig. 2: Warning tape- Quench tube

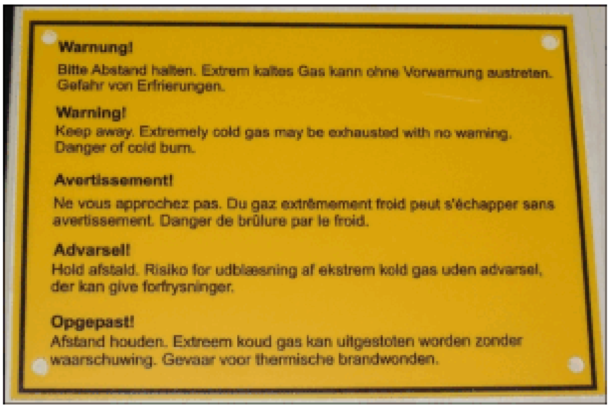
Example of new warning tape supplied with magnets. This will provide a consistent warning message.

2.4.3 External quench tube inspection

> Warning Signs
>
> The outlet warning sign must be in the National language.

Fig. 3: Warning sign example - Quench tube outlet

Fig. 4: Warning sign example - Quench tube outlet



Example of a new warning sign that provides a consistent message in several languages.

- Check the **warning sign** is fixed and clearly visible next to the **outlet** of the quench tube**.**
- Check the **outlet** of the quench tube for **obstructions** (e.g. bird's nest).
- Check for a **protective mesh** on the **quench tube outlet** and its integrity.
- Check a **protective rain cover** is fitted to the **vertical outlets**.
  - The cover must not be **damaged** or restrict the gas flow from the quench tube.

The **outlet** tube cannot be situated where, in case of a quench:-

- Helium gas can be drawn into an **air inlet.**
- Helium gas can enter **open windows**.

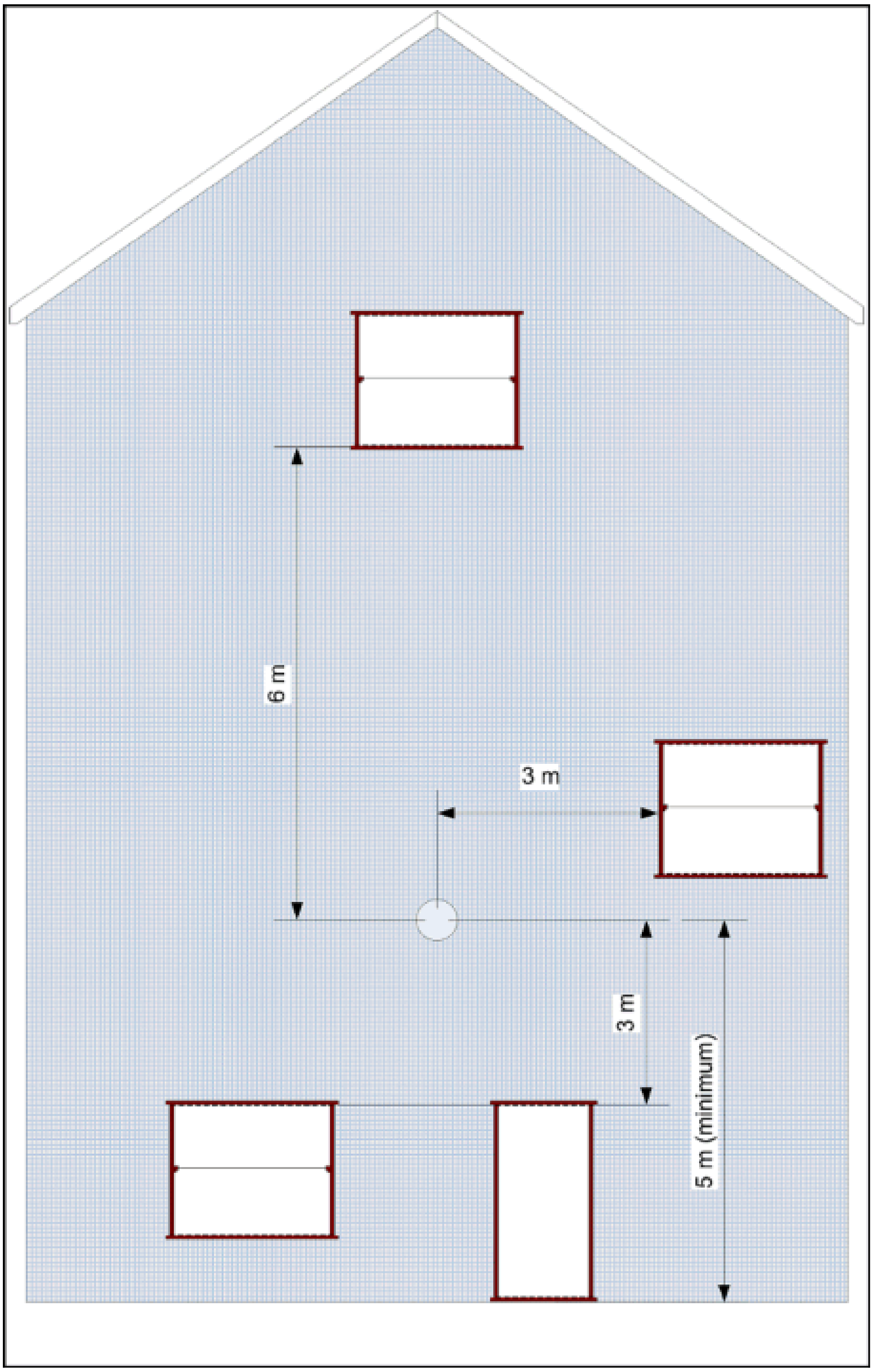
To avoid risk of **injury** from cold burns and asphyxiation, access to the quench tube must be restricted by:-

- 3 m on each side and below.(→ Fig. 5 Page 19)
- 6 m vertically above the exit.

> Helium is lighter than air, even when very cold. The cold gas must not be allowed to blow directly onto a window.

- Check that the restricted area has not changed.
- Check that **venting gas** does not represent a hazard to people. For example:-
  - Subsequent building of windows
  - Air conditioning system intake or exhaust pipes
  - Erection of new or temporary buildings
  - Temporarily storing containers.
- If windows are present in the restricted access area, they must be **sealed** and **permanently closed.** Means of opening the windows must be **removed**.
- Inform the operator to initiate corrective action **immediately** in case you find obvious **defects** or problems which could restrict the flow of gas from the quench **tube**.
- Record your findings in the **Protocol.**

Fig. 5: Quench tube - example of venting to the outside



2.5 Magnet

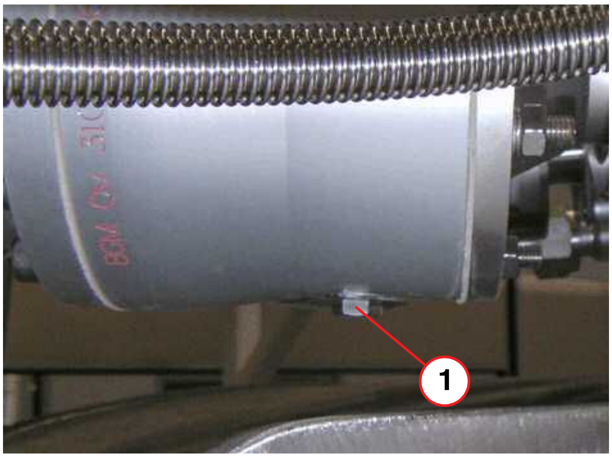
Required time: 50 min.

Required tools: 13 mm wrench, electronic helium leak tester, hand-held MSUP display unit.

2.5.1 Quench valve

SI Check for water in the quench valve

Fig. 6: Mk1 Quench Valve drain screw



(1) Brass drain screw

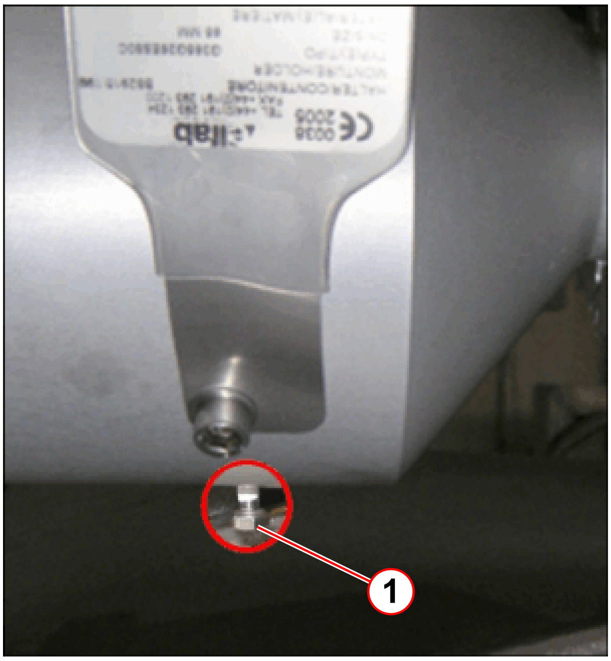
Mk1 Quench Valve

- Remove the turret covers.
- Remove the drain screw (1) and check for water
  - There **must not be any water** in the quench valve.
- **Refit** the drain screw - **DO NOT OVERTIGHTEN**.

> Always use the magnet service platform when working on the service turret.

> If water is present, determine the cause (e.g. damaged or missing protective rain cover, missing, quench tube insulation - leading to condensation). Advise the customer to initiate corrective action immediately.

Fig. 7: Quench valve drain screw



(1) Metal drain screw

Mk2 Quench Valve

- **Check that the drain screw has been removed.**
- Check the drain hole **remains unblocked** during each Service visit.

> For Static systems, the drain hole is permanently open - it allows water to drip freely from the quench valve.

> For Mobile Systems, fitted with high altitude venting, the drain hole must be closed (blanked with an M8 screw and plain washer).
>
> If the drain hole is left open, the high altitude venting is not effective.

2.5.2 Ice formation check

> Always use the magnet service platform when working on the service turret.

SI Check for ice formation at the magnet service turret and venting system

- The service turret and venting system should be free of ice formation (except after ramping up or filling).
  - » Empty out the water from the condensation reservoir located at the electronics side of the magnet.

2.5.3 Leakage check

SI Magnet system leak check

The magnet is fitted with a recondensing cold head and pressurization heater which maintains the pressure in the helium vessel at a constant **15.3 psi (A)**, providing there are no leaks in the venting system.

Gross leaks will show up as a **low magnet pressure**. Fine leaks may not show any change in pressure.

All leaks will show up as a slow steady drop in helium level.

> Insufficient cooling power of the cold head could also result in helium loss. If the heater average power = 0W and magnet pressure ~ 16 psi (A), then check the cold head performance).

Leak checking must be performed with an **electronic leak detector**, capable of detecting leaks, at least as small as **1 x 10^{-4} mbar l/s.**

Fig. 8: Edwards Leak detector

- Use the Edwards hand-held leak detector (part no **10614850**).

  This is the **recommended equipment** for leaks **greater than 1 x 10^{-4} mbar l/s.**

NOTE: The hand-held device is calibrated in **ml/sec**:-

1 x 10^{-4} mbar l/s **=** 1 x 10^{-4} ml/sec at normal pressure and temperature.

A **leak check** must be performed if the magnet has **low pressure** and **helium loss.**

- Check the **magnet pressure**, and the **helium level history** in the Service Software >**Magnet & Cooling**.
- If the magnet pressure is out of tolerance, or helium loss has been detected:
  » Check all **joints** and **seals** on the **high-pressure side** of the magnet venting.

2.5.3.1 Helium level check

If the magnet has **low pressure** or there is evidence of slow helium loss, the following checks must be carried out:-

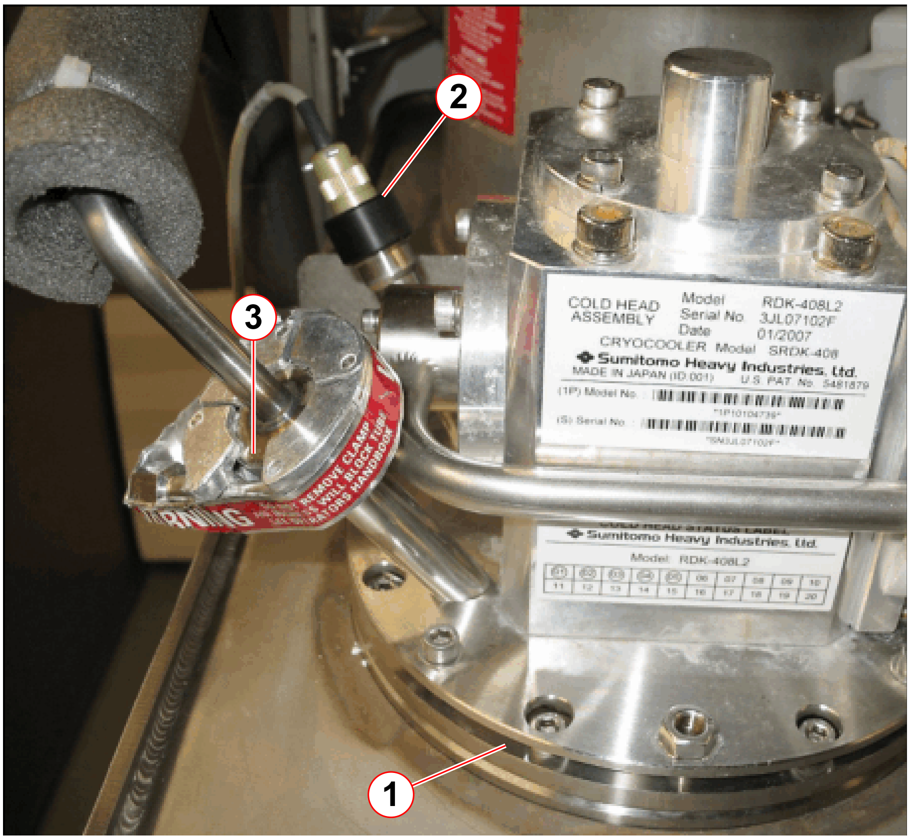
> If no external leak is found, but the system pressure is low, it is possible that one of the internal valves is leaking. this can be caused by a cracked or broken burst disk, or a broken O-ring.

> The magnet must be RUN TO ZERO before checking inside the quench valve or auxiliary vent.

Cold head

> The cold head vent line is to remain connected with the hand-valve CLOSED during normal system operation.

Fig. 9: Cold head seals - leak check



(1) Cold head flange seal O-ring
(2) Cold head temperature sensor & O-ring seal
(3) Cold head vent line NW16 seal (high load)

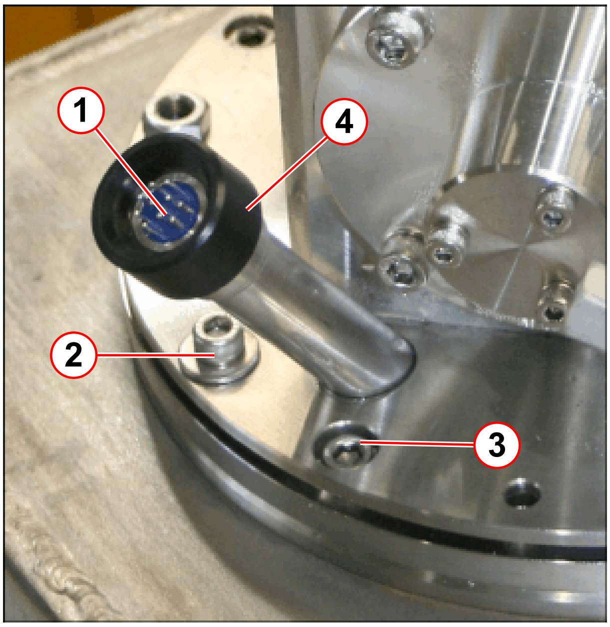
- The **cold head** is sealed in position by an **O-ring.** (→ 1/Fig. 9 Page 23)
  - leak test around **this point.**

- If there is a **leak** around the **cold head seal** check the **torque** of the cold head **screws**.

> ℹ If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5Nm.
>
> If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

- Check the **cold head vent line seals** at both ends.
  - leak checking around the **NW16 connector.**

Fig. 10: Cold head 10 pin seal

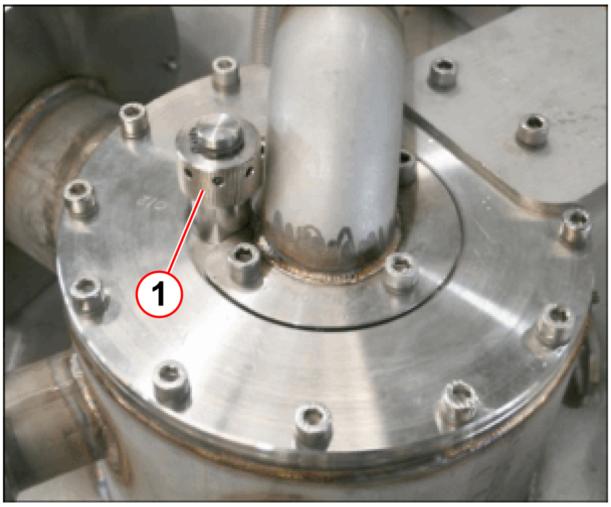


(1) Leak check point
(2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
(3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
(4) Black sealing cap

- Remove the **temperature sensor cable**.
- Leak check the **10 pin seal.**

Syphon port

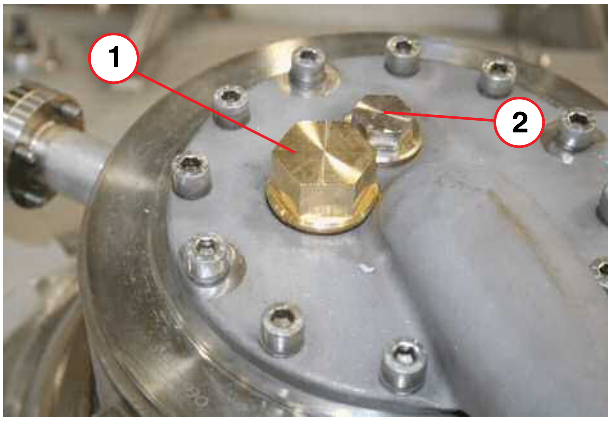
Fig. 11: Syphon entry port



(1) Syphon blanking plug

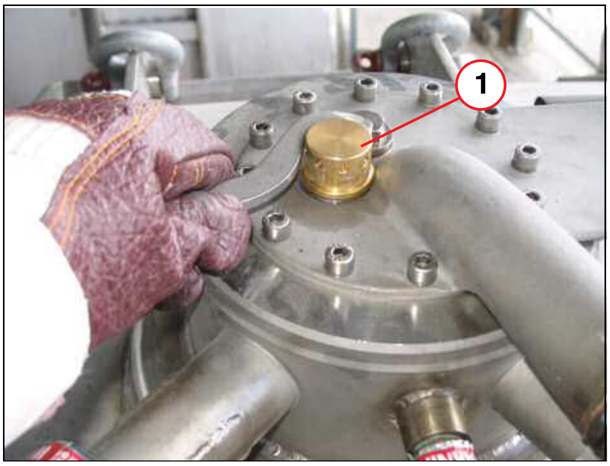
- Check the **helium fill port** and other seals on the **turret top plate** with a helium leak detector.

Fig. 12: Mk2 Syphon port entry



(1) Syphon blanking plug
(2) Sight glass

Fig. 13: Mk2 Syphon port port

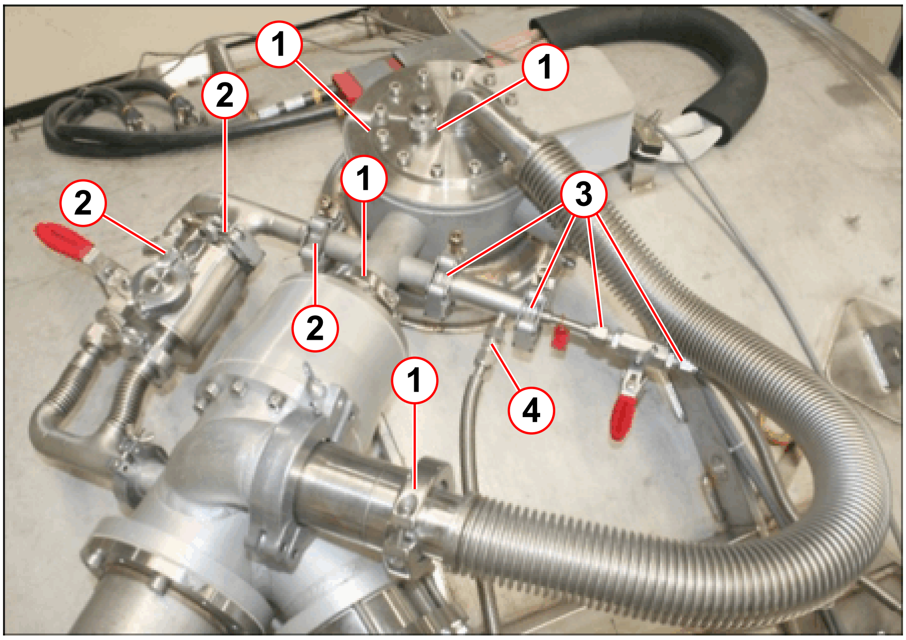


(1) Syphon blanking plug

> If a leak cannot be found, but there is a slow, steady helium loss, report to your local Uptime Centre for repair by a specialist.

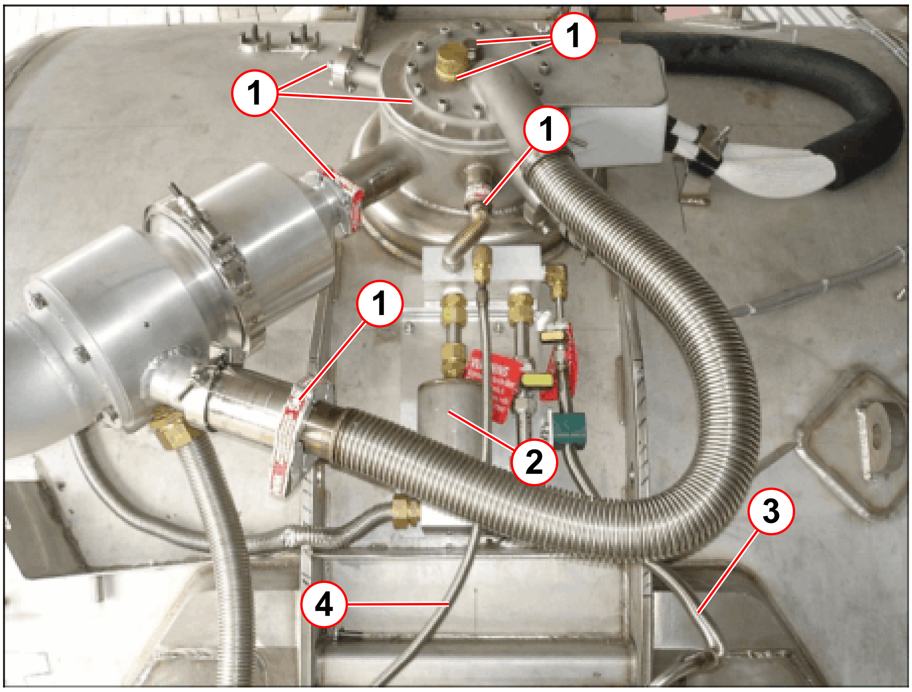
Turret & Venting

Fig. 14: OR105 Mk1 - Turret and venting leak checks (high pressure)



(1) Turret top plate, syphon port and O-rings
(2) 16 psi (A) vent valve
(3) Cold head line
(4) Pressure gauge line

Fig. 15: OR105/122 Mk2 - Turret and venting leak checks (high pressure)



(1) Turret top plate, syphon port and O-rings
(2) 16 psi (A) vent valve
(3) Cold head line
(4) Pressure gauge line

- Leak check all the **clamping points** on the **high-pressure side**. If there is a leak then, it could make an existing **leak worse.**

Do not use any form of vacuum grease on 'O-ring seals.

Wipe clean and check for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

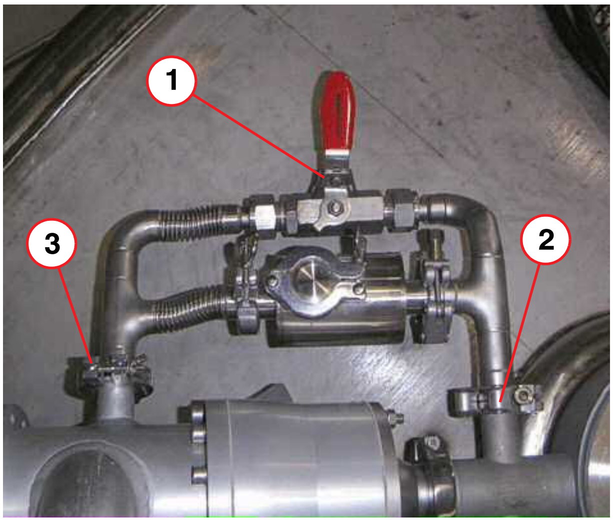
Replace a high load seal only with another identical high load seal. Generally high load seals can only be used ONCE for a guaranteed leak tight seal.

16 psi (A) Vent valve assembly

- **Leak check** the 16 psi (A) Vent valve assembly.

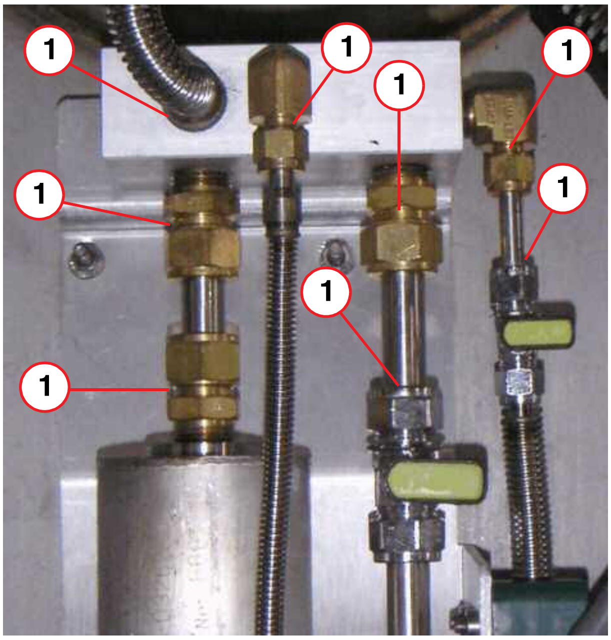
Mk1 Vent valve assembly	Mk2 Vent valve assembly

Fig. 16: OR105 Mk1 Vent valve assembly



(1) Bypass valve
(2) NW25 High pressure clamp (turret)
(3) NW25 Low pressure clamp (manifold)

Fig. 17: Mk2 Venting 16 psi (A) Valve Assembly



(1) Leak test points

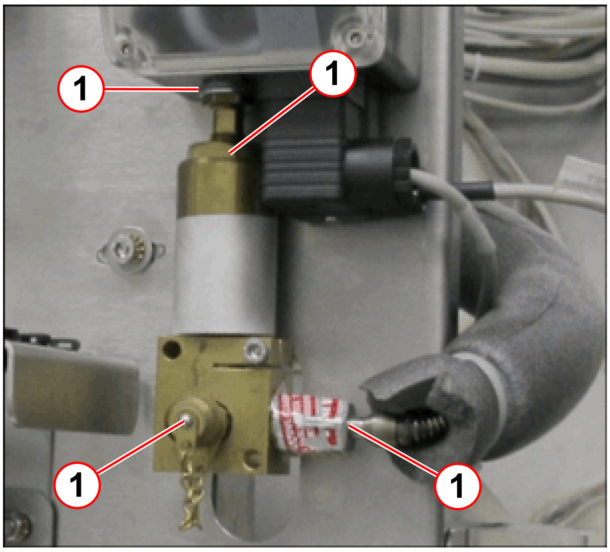
- Leak check the 16 psi (A) valve **high-pressure** assembly.
- For leaks found at pipe couplings, first try tightening the clamp or securing screws. **DO NOT OVER-TIGHTEN!**
- If tightening does not solve the leak, the **complete** vent valve assembly must be replaced.
- If leaks are found at **connections** into the inlet aluminium block, the **complete** assembly **must be replaced**.

Pressure gauge assembly

> The WIKA Pressure switch transducer is NOT directly interchangeable with the MK1 Manocomb switch. Contact Magnet HSC for assistance.

Manocomb leak check points

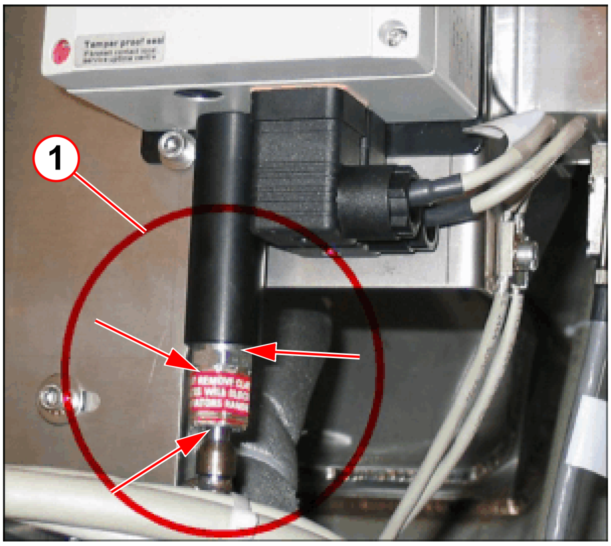
Fig. 18: Manocomb - leak check points



(1) Leak check points

WIKA leak check points

Fig. 19: Pressure switch assembly - leak checks



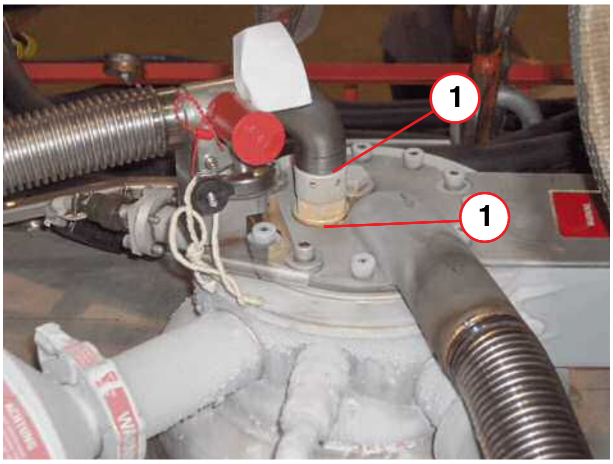
(1) Leak check points

- **Leak check** the **high-pressure** points.

Contact your local **Uptime Service Center** if further leaks are suspected.

Mobile systems

Fig. 20: Mk2 Mobile syphon (permanent fitment)



(1) Leak test points

For mobiles fitted with a **permanent syphon**, the following checks **must** be carried out.

- Check the **syphon entry** is tight. Leak check if a leak is suspected
- Check that the **syphon end cap** is fitted and is leak tight
- Contact your local **Uptime Service Centre** if further leaks are suspected.

All static system checks (i.e. low magnet pressure), will also apply to mobiles.

Do not use any form of vacuum grease on O-ring seals. Vacuum grease dries over time and can cause a seal to leak.

Wipe clean and check for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.

Replace used high load seals. The high load seals (copper (CU) or Heli-flex types) must be replaced with a new one.

Heli-flex seals are preferred, and should be used to replace the copper seals.

Generally high load seals can only be used ONCE for a guaranteed leak tight seal.

2.6 Patient fan

SI Checking the air filter of the patient fan

Required time: 10 min

Required tools and parts: 5mm Allen key, air filter in case of pollution or damage.

- Switch off the patient fan on the control unit.
- Remove the lower service cover.
- Remove the top service cover.

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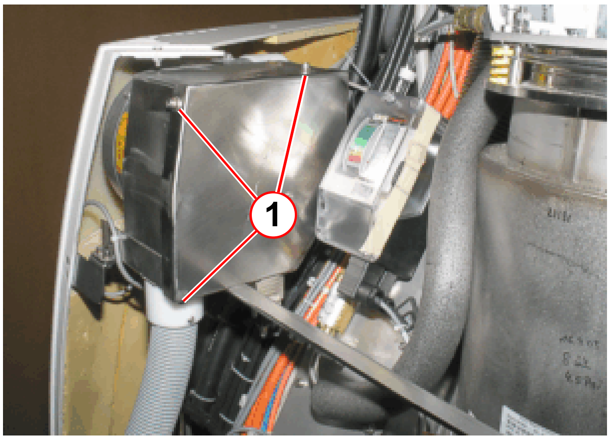
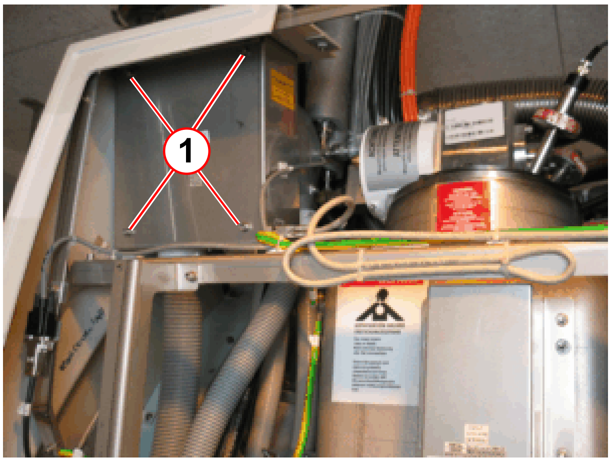
Fig. 21: Patient fan



(1) 5 mm allen screws

Fig. 22: Patient fan



(1) 5 mm allen screws

- Remove the cover of the patient fan.
- Loosen the Allen screws (1).

NOTICE

Ferromagnetic parts.

The filter grid and the wing nut are slightly ferromagnetic.

» **Hold the parts tightly. Handle the filter grid and wing nut with care.**

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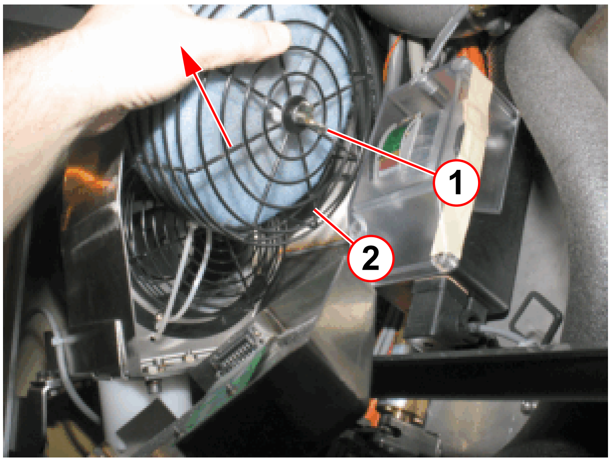
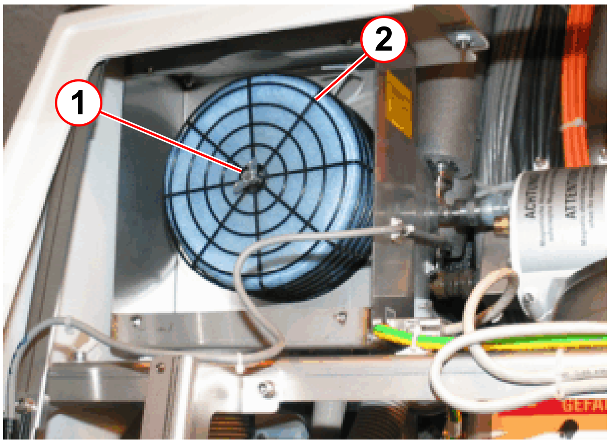
Fig. 23: Patient fan



(1) Wing nut
(2) Filter grid

Fig. 24: Patient fan



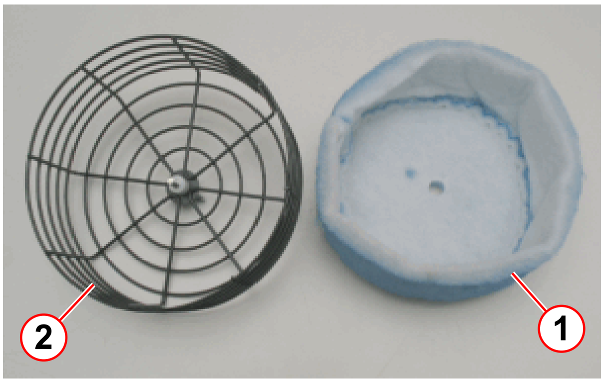
(1) Wing nut
(2) Filter grid

- Loosen the wing nut.
- Remove the filter grid (for Avanto; remove the filter grid by slightly tilting and moving upwards).

The filter grid is protected with a cable tie.

- Cut the cable tie.

Fig. 25: Patient fan air filter



(1) Air filter
(2) Filter grid

- Replace the air filter (1).

- Install the new air filter with grid on the fan body in reverse order.
- Switch on the patient fan on the control unit.
- Install the top service cover.
- Install the lower service cover.

2.7 User documentation

Required time: 5 min

Required tools: n.a.

SI User documentation available and legible

- Ensure that the System Manual is available, legible and valid (e.g. software version).

2.8 User icons, button labels

Required time: 5 min

Required tools: n.a.

SI Checking user icons and button labels

- Inspect the button labeling and user icons for legibility/recognizability (for example “symbol keypad”, “**Magnet Stop button**”, “Patient Table Stop”, etc.).

2.9 Identification of the controlled access area

2.9.1 Identification of the controlled access area (0.5 mT area)

Required time: 10 min

Required tools: n.a.

Refer to the Spare Parts Catalogue for the order number for the set of warning signs.

SI Identification of the controlled access area (0.5 mT area)

Fig. 26: Warning sign "Strong Magnetic Field"

Fig. 27: Warning sign " Cardiac Pacemaker"

Prior to initial Startup, the project manager defines the 0.5 mT area.

- Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone. The signs have to be posted in a way that it is visible even when the door to the controlled area is opened.

SI Checking the warning signs (SIEMENS)

SI Checking the warning signs (customer signs cover all safety aspects as listed on the SIEMENS signs)

The fringe field extends three-dimensional beyond the magnet and can be reduced through shielding. The field lines represent the ideal flux density distribution in air, which may be distorted by steel reinforcement in buildings.

2.9.2 Fringe field Avanto

Fig. 28: Fringefield Z / Y

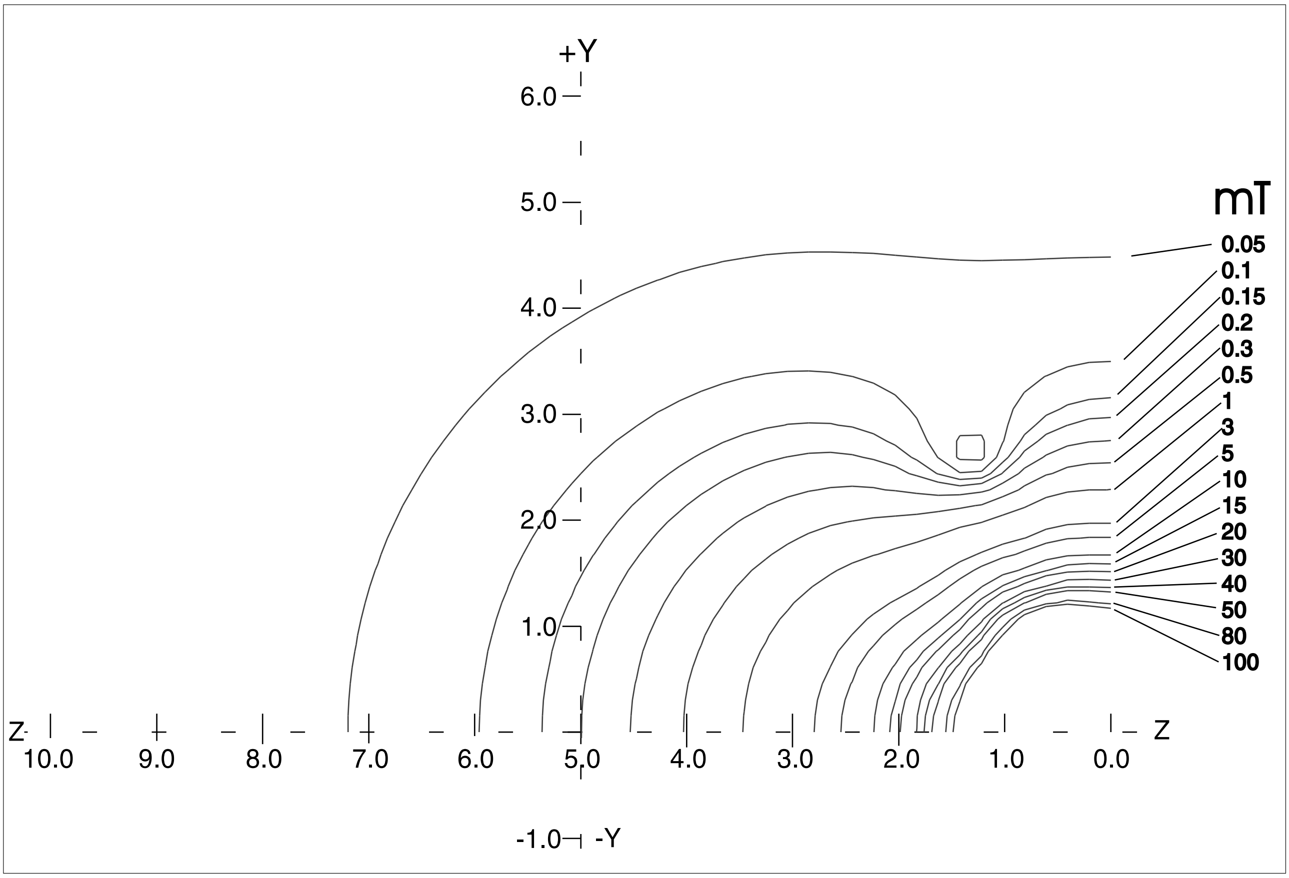


Fig. 29: Fringefield Z / Y

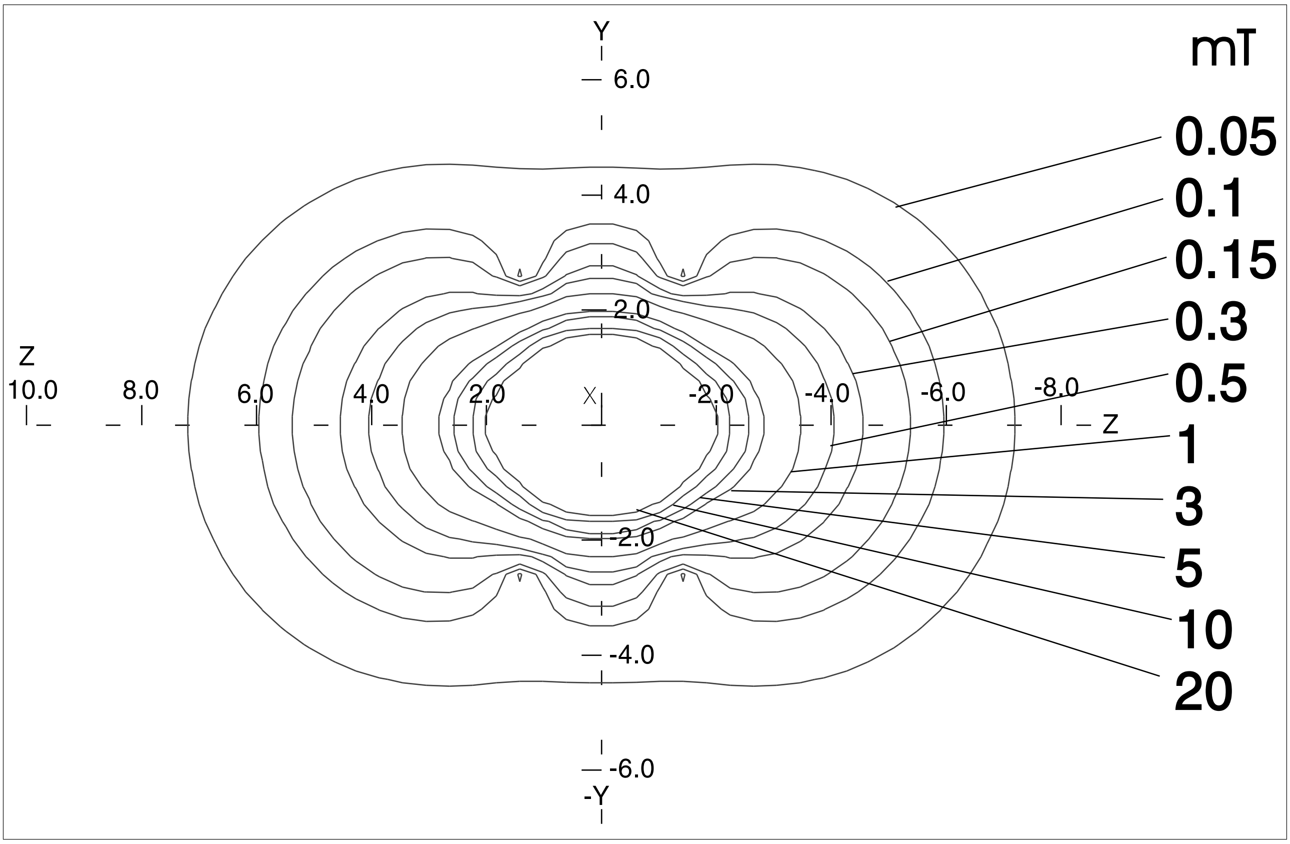
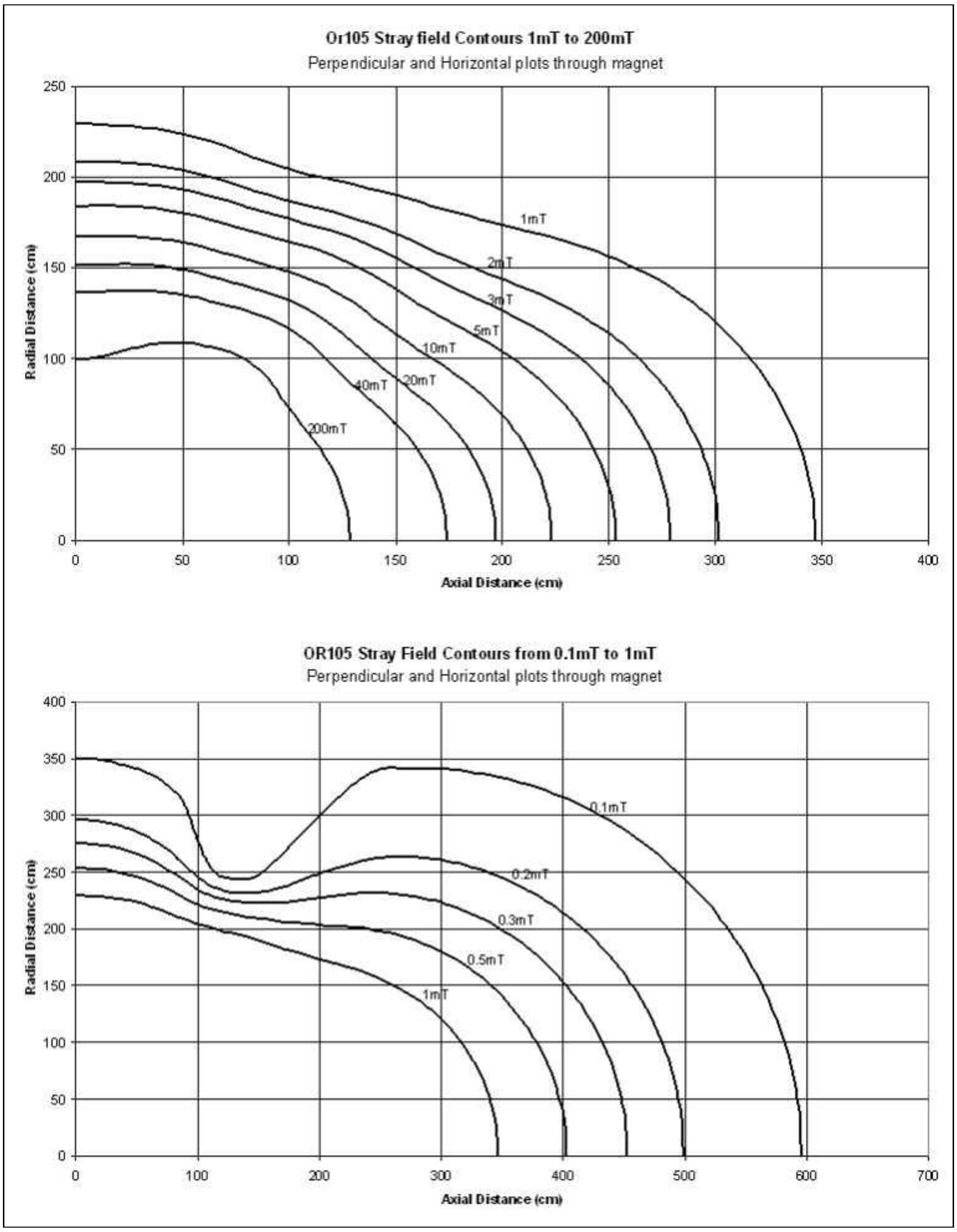


Fig. 30: Fringe field distribution



Tab. 6 Fringe field distribution Avanto

Fringe field	Distance from the magnetic center in direction of		
	X-axis in [m]	Y-axis in [m}	Z-axis in [m]
40 mT	1.4	1.4	1.8
20 mT	1.6	1.6	2.0
10 mT	1.7	1.7	2.3
5 mT	1.9	1.9	2.6
3 mT	2.0	2.0	2.8

1 mT	2.3	2.3	3.5
0.5 mT	**2.50**	**2.50**	**4.00**
0.3 mT	2.8	2.8	4.5
0.15 mT	3.2	3.2	5.4
0.1 mT	3.5	3.5	6.0
0.05 mT	4.5	4.5	7.2

2.9.3 Fringe field Espree

Fig. 31: Fringe field distribution chart (Y/Z)

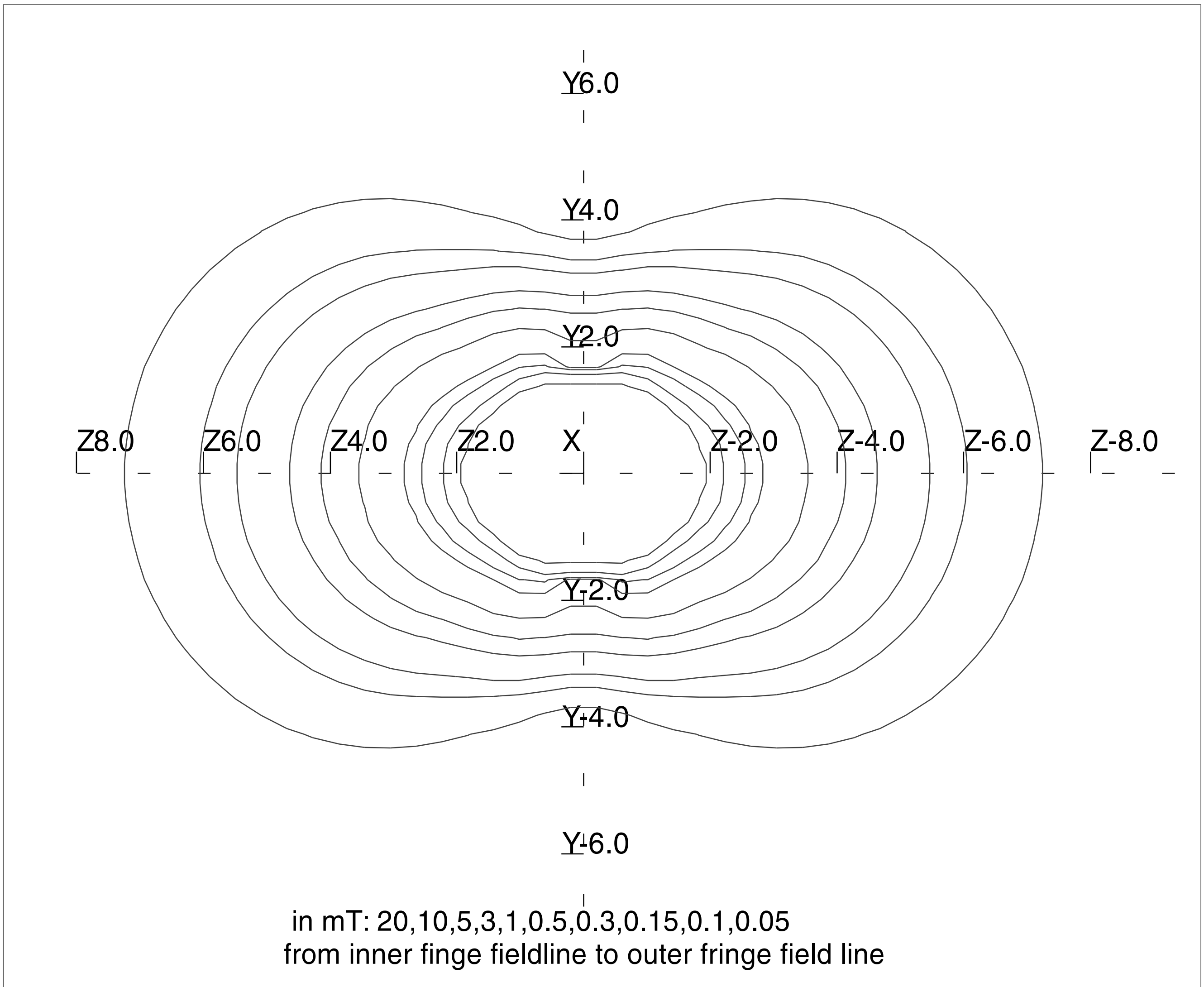
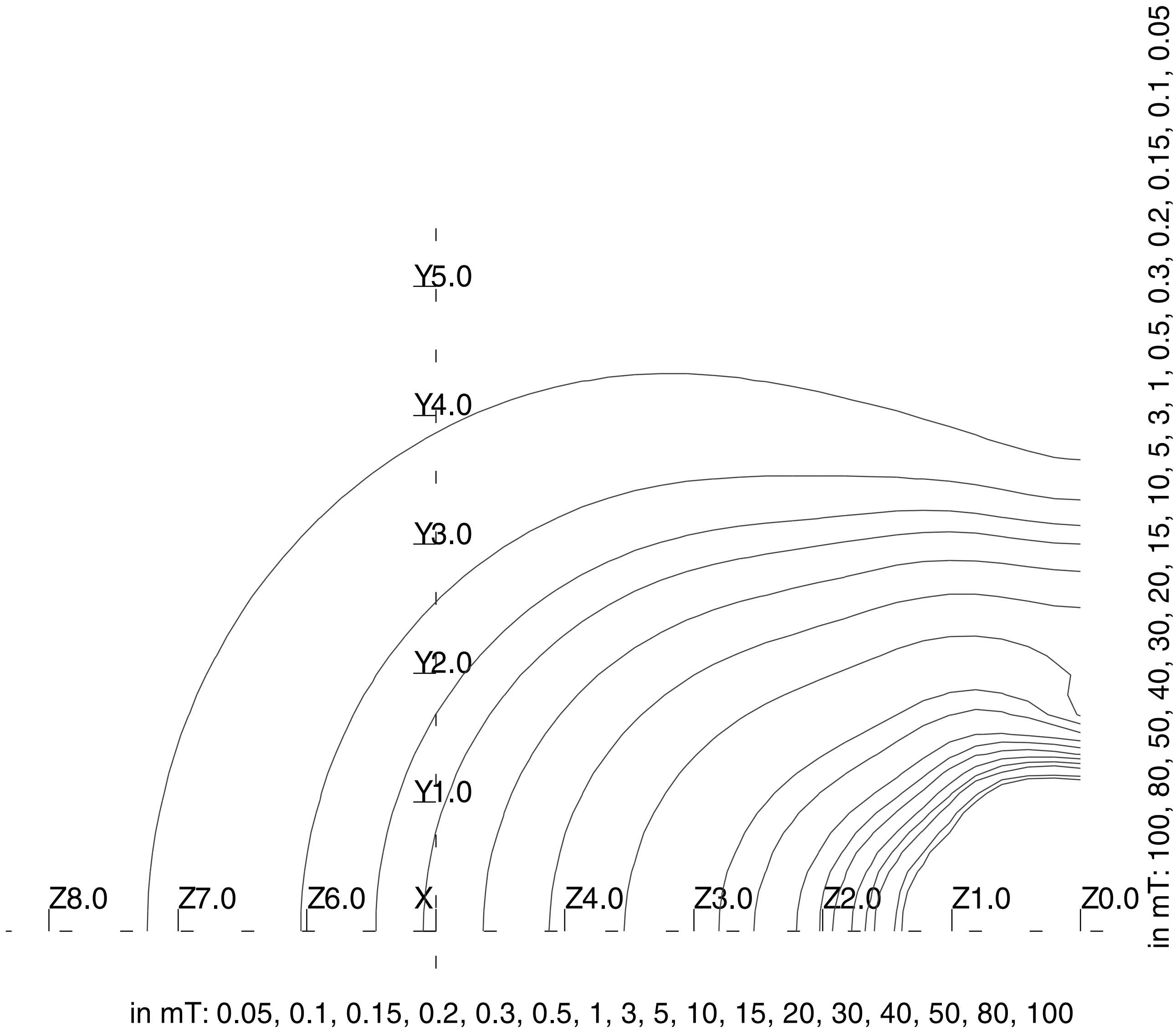


Fig. 32: Fringe field distribution chart (Z/Y)



Tab. 7 Fringe field distribution Espree

Fringe field	Distance from the magnetic center in the direction of		
	the X-axis in m	the Y-axis in m	the Z-axis in m
40 mT	1.3	1.3	1.7
20 mT	1.4	1.4	1.9
10 mT	1.6	1.6	2.2
5 mT	1.7	1.7	2.5
3 mT	1.9	1.9	2.8

1 mT	2.3	2.3	3.5
0.5 mT	**2.5**	**2.5**	**4.0**
0.3 mT	2.9	2.9	4.6
0.15 mT	3.3	3.3	5.4
0.1 mT	3.4	3.4	5.8
0.05 mT	4.3	4.3	7.2

2.10 RF room door

Required time: 5 min

Required tools: n.a.

2.10.1 Door lock

SI Checking the door lock of the RF room

1. The locking and unlocking mechanism has to function properly.
2. The door handle or knob has to be securely fastened.

In case of a defect (e.g. defective unlocking mechanism, door handle,), inform the customer to initiate corrective action immediately.

2.11 Warning Signs

2.11.1 Laser warning labels

Required time: 3 min

Required tools: If warning label needs replaced , order p.n. 7546430

SI Checking the laser warning labels

- Ensure that the correct warning labels are affixed to the cover.

  If the option “control unit magnet rear” is installed, ensure that the correct warning labels are affixed to the cover.

  » The labels must be legible.
- Replace missing labels immediately.

Fig. 33: e.g. Avanto magnet cover / laser label

2.11.1.1 Label configuration for the US Certification label, warning logo type and aperture label

Fig. 34: Laser label configuration USA

(1) Aperture label, warning logo type

(2) Certification label

2.11.1.2 Label configuration for all other countries in local language or alternative language.

Fig. 35: Laser label

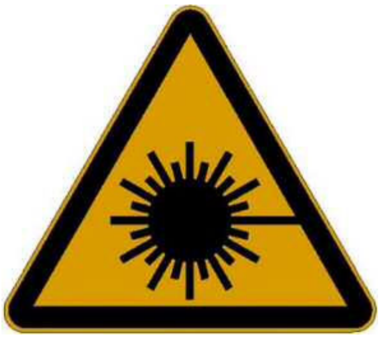


Fig. 36: Laser label 1 (new version)

Fig. 37: Laser label 1 (old version)

2.11.2 Warning signs of the Patient Table

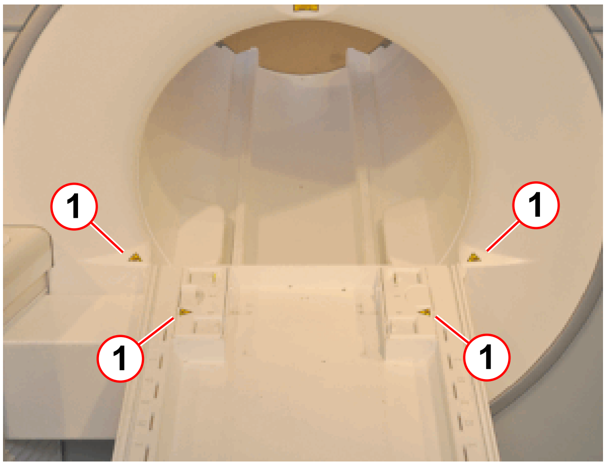
Required time: 3 min

Required parts and tools: Sign (➔ Fig. 38 Page 45) if missing. Sign is part of the label set part number 7546430

Fig. 38: Warning sign 'bruising hazard'

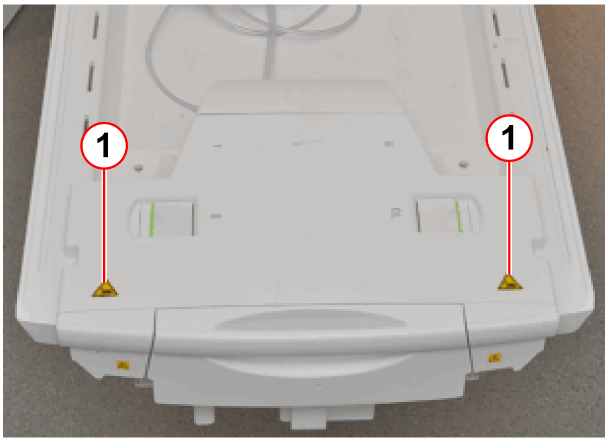
SI Checking the warning signs of the patient table

Fig. 39: Patient table



(1) Warning sign 'bruising hazard'

Fig. 40: Patient table



(1) Warning sign 'bruising hazard'

- Ensure that the warning labels are affixed to the patient table and to the magnet bore .
- Replace missing labels immediately.

2.11.3 Hearing protection sign

Required time: 3 min

Required tools or parts: if missing hearing protection sign. The hearing protection sign is part of the warning sign sets, please refer to SPC.

SI Checking the hearing protection sign

- Ensure that the hearing protection sign has been posted at all entrances to the examination room.

Fig. 41: Hearing protection sign

- Install missing signs immediately

2.12 End of General Tests

End of section.

3.1 Measurement of the ground wire

3.1.1 Measurement of the system ground wire (every 2 years)

Required time: 30 min

Required tools: Ground test meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective conductor test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results!

> **WARNING**
>
> **The ground test meter is ferromagnetic! Do not bring it into the examination room while the magnet is at field!**
>
> **Failure to comply may cause the ground test meter to become a projectile due to the magnetic force, resulting in serious injury or even death.**
>
> » **Perform protective conductor tests using measurement cables of a sufficient length or after the magnet has been ramped down.**

» The protective conductor resistance must not exceed **0.3 ohm**.

- Use the **ground test meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution panel and a non-insulated metallic part of the:
- ACC MR electronics cabinet (ground bolt - power distribution box)

SI ACC - ground bolt - line distribution box (limit value ≤ 0.3 Ω)

- RF-filter plate (ground bolt)

SI RF-filter plate - ground bolt (limit value ≤ 0.3 Ω)

- Magnet (magnet housing)

SI Magnet - magnet housing (limit value ≤ 0.3 Ω)

- RF components on the magnet (mounting plate)

SI RF components on the magnet - mounting plate (limit value ≤ 0.3 Ω)

- Patient table (screw at table underside)

SI Patient table - screw at table underside (limit value ≤ 0.3 Ω)

- Magnet cold head compressor (housing)

SI Magnet cold head compressor - housing (limit value ≤ 0.3 Ω)

- MRC host (housing screw)

SI MRC-Host - housing screw (limit value ≤ 0.3 Ω)

- Option SEP (metal frame)

SI Option SEP - metal frame (limit value ≤ 0.3 Ω)

- Option IFP (metal frame)

SI Option IFP - metal frame (limit value ≤ 0.3 Ω)

- Intercom operating unit (housing screw)

SI Intercom operating unit -housing screw (limit value ≤ 0.3 Ω)

- Intercom (housing screw)

SI Intercom - housing screw (limit value ≤ 0.3 Ω)

- Power distribution for the console components (ground screw)

SI Power distribution for the console components - ground screw (limit value ≤ 0.3 Ω)

- MRC TFT monitor (housing screw at the bottom (→ Fig. 42 Page 51))

SI MRC TFT monitor - housing screw at the bottom (limit value ≤ 0.3 Ω)

- All other options such as **UPS, patient monitor**, **in-room MRC, Digicam,** etc.

> Ground wire measurement for CCD-Vario-Camera of PAMO5 is not necessary as it is supplied with low voltage.

SI Option 1 (limit value ≤ 0.3 Ω)

SI Option 2 (limit value ≤ 0.3 Ω)

SI Option 3 (limit value ≤ 0.3 Ω)

> Enter N.A. in the measured value field if the component / option is not present.

- Enter the values measured for each component in the Protocol.

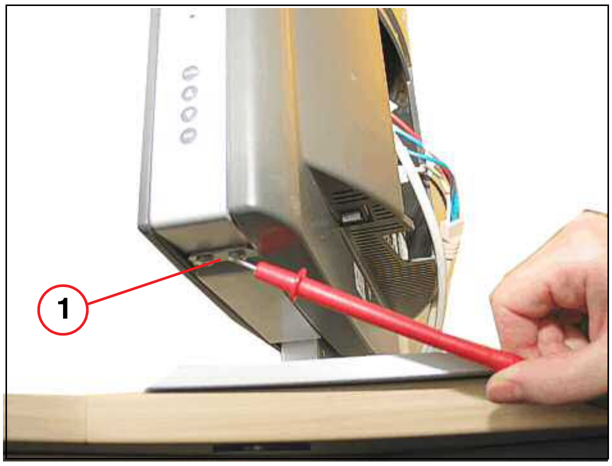
⚠ WARNING

Exceeding threshold values!

In case of electrical fault, exceeding a threshold value may result in death or serious injury.

» **Test to determine what exceeds the threshold value. Implement actions immediately to reduce the value prior to continuing work.**

Fig. 42: TFT monitor



(1) Measuring point

> **NOTICE**
>
> **Option: In-room MRC**
>
> **Do not touch the trackball with the probe of the ground test meter.**
>
> **The high current used for measuring the resistance may damage the small contact springs of the trackball during the measurement.**
>
> » **To perform the measurement, use the uncoated screws at the back of the top cover of the In-Room MRC.**

3.1.2 Measurement of the MRSC ground wire (annually)

Required time: 10 min

Required tools: Ground test meter

Use the ground test meter to measure the ohmic resistance between the ground pin at the plug of the power distribution for the MRSC components and a non-insulated metallic part of the:

- MRSC host (housing screw)

  SI MRSC Host - housing screw (limit value ≤ 0.3 Ω)

- TFT monitor (housing screw at the bottom)

  SI TFT monitor housing screw at the bottom (limit value ≤ 0.3 Ω)

- Option desktop MOD, external

  SI Option external desktop MOD (limit value ≤ 0.3 Ω)

  » The protective conductor resistance must not exceed **0.3 ohm**.

- Enter the values measured for each component in the Safety-Related Tests Protocol.

3.2 End of Electrical Tests

End of section.

4.1 TALES

Required time: approx. 70 min

Required tools and parts:

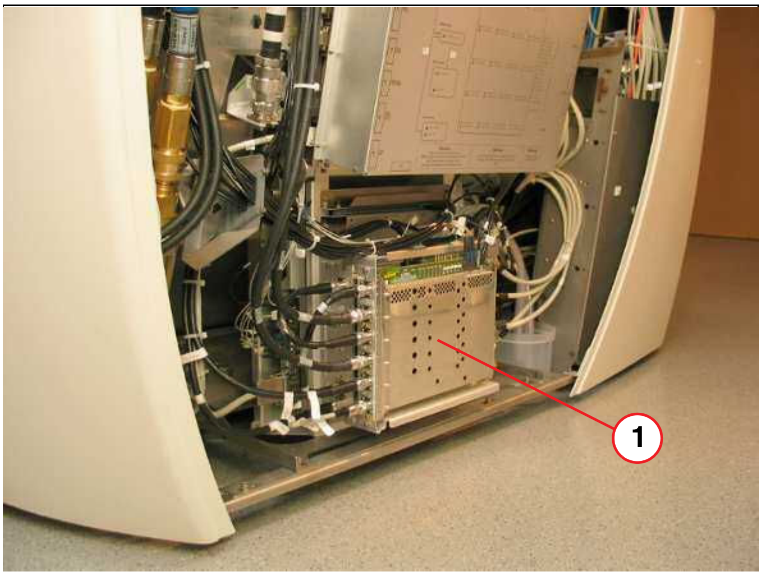
- TALES
- Screwdriver
- ESD kit
- only if MNO option is installed: Service Plug kit (07715720)
- Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration,
- BNC 2m Service cable for step TU/ Tuning Calibration (8733479)

Prerequisite: Service access covers are removed, system is switched off

SI Replacing TALES (every 2 years)

> Do not replace the assembly for a new installation
>
> The initial replacement in this case occurs in 2 years.

Fig. 43: Tales



4.1.1 Preliminary steps

1. Switch the system to standby mode by selecting **"System,"** "Control" in the main menu.
2. Click the task card "MR Scanner."
3. Click "stand-by."
4. Switch off ACC/F24.

5. Remove the left service access cover from the magnet.

4.1.2 Disassembling / Assembling

1. Disconnect all cables from the TALES.
2. Remove the old TALES and install a new one (use ESD kit for electrostatically sensitive devices).
3. Reconnect all cables. Firmly tighten the N- connectors by hand (1 - 2 Nm).
4. Switch on ACC/F24.
5. Click **System** > **Control** in the main menu.
6. Click the task card “MR Scanner.”
7. Click “System ON.”

4.1.3 Adjustments:

1. Start the Service platform.
2. Open the Tune-up platform by pressing the corresponding button.
3. Perform the following Tune-up steps:

   a) Tuning Calibration
   b) BC Tuning
   c) RF Characteristic
   d) RF Characteristic LC (only Avanto and, if multi nucleus spectroscopy option is installed)
   e) Coil Power Losses or BC Power Losses (VD13)
   f) Abs. Rec. Path Calib. or BC Image Brightness (VD13)

> Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the ‘Help’ button in the header bar of the corresponding platform.

4.1.4 Final checks:

1. Open the Quality Assurance platform by pressing the corresponding button.
2. Perform the following Quality Assurance steps:

   a) Abs. Rec. Path Calib. Check or BC Image Check (VD13)
   b) BC Tuning Check
   c) RF Verify
   d) RF Verify LC (only Avanto and, if multi nucleus spectroscopy option is installed)
   e) Coil Power Losses Check or BC Power Losses Check (VD13)

f) Expert mode > RF Path Attenuation Check
g) If the system has a local TX coil installed, perform QA Coil Check for this specific coil.

3. Label the return spare part “Tales": safety maintenance.”
4. Return the replaced TALES to Erlangen.
5. Install the left service access cover.

4.2 End of Mechanical Tests

End of section.

5.1 Emergency buttons

Required time: 15 min

Required tools: n.a.

SI Emergency shutdown buttons

Prerequisites:

The system contactor is switched on.

The system is not booted (system shutdown or press **F2** during boot).

- Press the **emergency shutdown button** (EPO).
- Check that the system contactor in the electronics cabinet is de-energized.
- Check that all system components are voltage-free (e.g. imager, host computer).
- If a UPS (Uninterruptible Power Supply) is installed, it also must be switched off by pressing the EPO.
- Check that the **"Line"** LED at the alarm box is off.
- If there is an audible signal, press the **Alarm Silence** button on the alarm box to stop the signal.

  » If the RF-room has an electric or pneumatic door, it has to be possible to open it in the event of a power error.
- Check if the RF-room door can be opened when the system is "powered off" .

Release the emergency shutdown button and reactivate the system by switching on the main power.

- Perform this test on all emergency shutdown buttons in the equipment room, control room, and the examination room.

5.2 Gradient Supervision Avanto

Required time: 40 min

Required tools: Smoke wick material number 10684293 or test spray material number 10684091, pocket lighter, and screwdriver

Prerequisites: The display of the aspiration smoke detector shows OK (→ 1/Fig. 52 Page 63). System is on.

- Remove the following cover sections; please refer to Replacement of Parts > System > Cover
  - Left rear ring segment
  - Right rear ring segment
  - Top rear cover

WARNING

Accumulation of oxygen during helium filling.

The use of burning smoke wicks during helium filling could result in an open fire in the area of the magnet service turret.

» **Do NOT perform a smoke test during helium filling!**

SI Checking the gradient supervision

- Take the smoke wick out of the box.

Fig. 44: Smoke wick in the box

NOTICE

On-site smoke detectors

Avoid false alarm of the on-site smoke detector!

» **Check if there is an on-site smoke detector. The smoke or test gas must not enter the on-site smoke detector.**

» **Keep the smoke or test gas away from on-site smoke detectors.**

» **Keep the smoke time as short as possible to minimize smoke concentration in the air.**

- Light the tip of the smoke wick and allow it to burn for 10 - 15 seconds. Then blow out the flame.

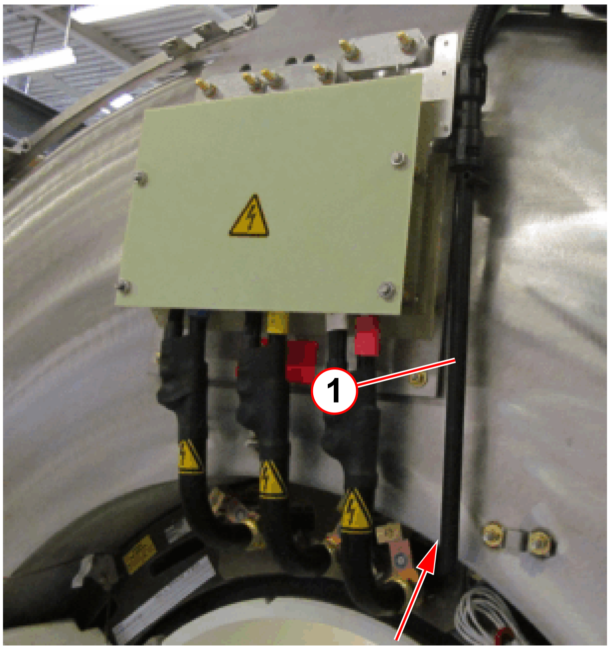
Fig. 45: Burning smoke wick

(1) Plastic handle

» The smoke wick provides a stable and continuous trail of smoke.

- Hold the smoke wick behind the rear cover for 5 seconds so that the smoke aspirated and transported to the detector (→ 1/Fig. 46 Page 59).

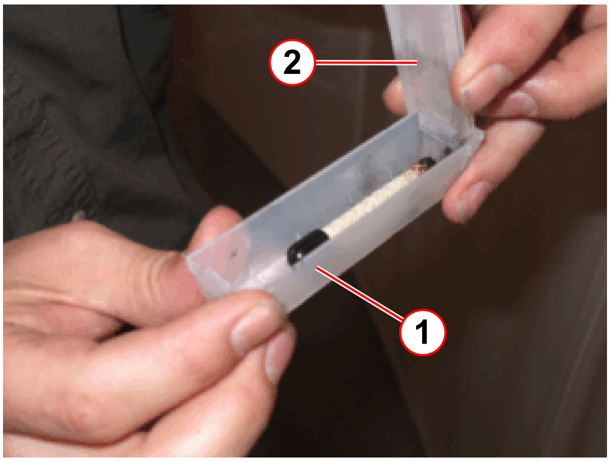
Fig. 46: Magnet service end / gradient coil connection



(1) Intake manifold (The assembly of the manifold can vary)

- After use, immediately place the smoke wick into the empty box and close the lid carefully.

Fig. 47: Extinguishing the smoke wicks



(1) Box
(2) Lid

» This will extinguish the smoke wick after approximately 15 seconds.

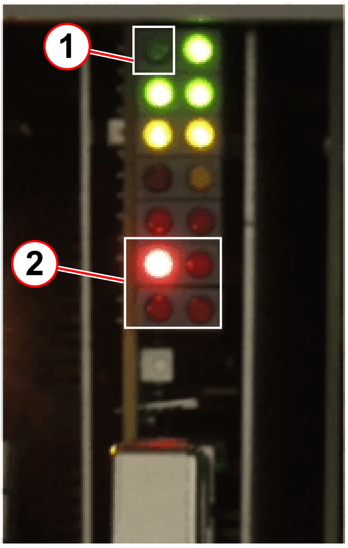
Alternatively, the test spray can be used. Spray the test gas dosed toward the intake manifold for 5 seconds. Dosed means in this context to press the spray button not completely. Distance between spray and intake manifold should be approximately 15-20 cm.

Fig. 48: Test spray

Expected test results:

- The smoke is being sucked (driven by 2 small fans) from the gradient manifold to the aspirating smoke detector. This will take up to 40 seconds (depending on the hose length), then the following results are expected simultaneously:
  » The red “FIRE” lamp on the VLM device must light up.
  » The gradient modulator in the GPA / GSSU is disabled (red LED’s, error code 8).

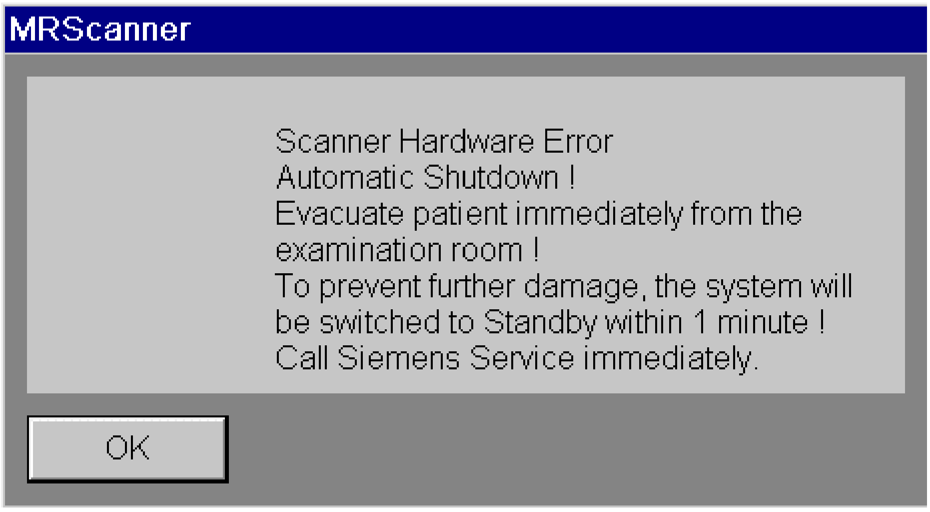
Fig. 49: Gradient modulator D40/ D41

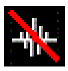


(1) Modulator off
(2) Error code 8

» An alarm message appears at the main console. (can also be checked in the error log).

Fig. 50: Pop-up "Automatic Shutdown"



and the scanner icon shows error  .

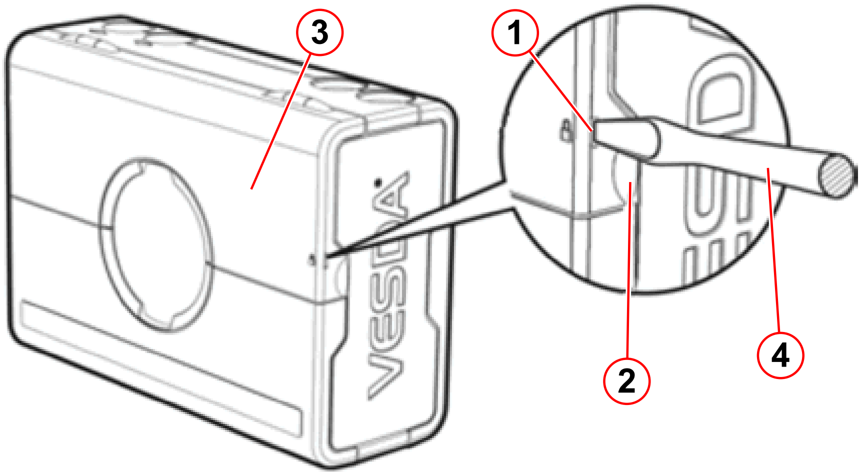
1 minute later,

» The system must be switched to stand-by mode automatically.

Reset the system:

- Press a screwdriver into the security tab (→ 1/Fig. 51 Page 62) to open the service access cover with the finger clip (→ 3/Fig. 51 Page 62).

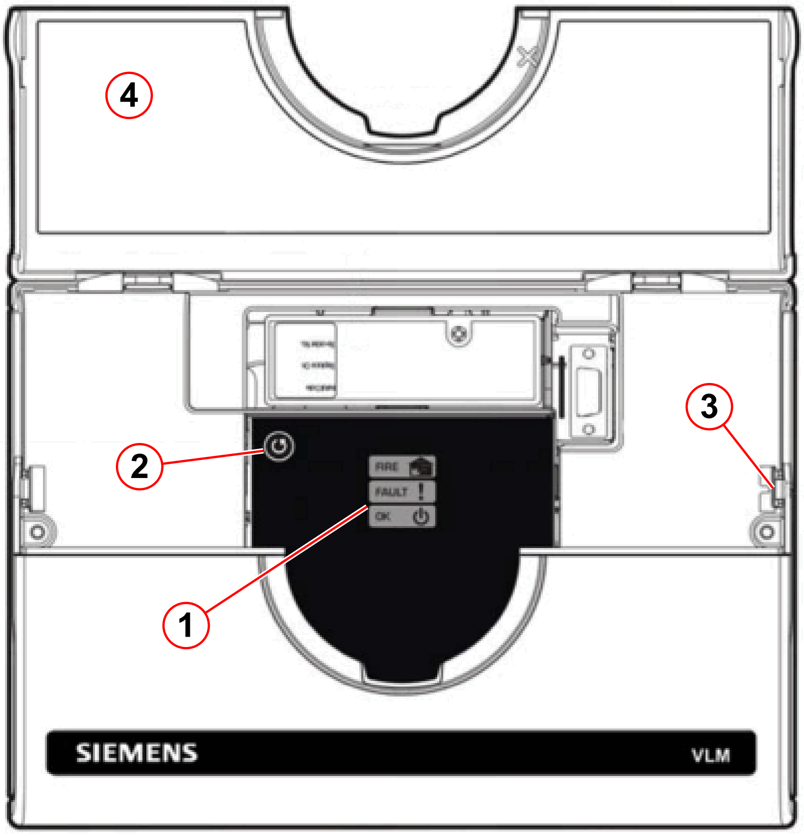
Fig. 51: Aspiration smoke detector ASD



(1) Security tab
(2) Finger clip
(3) Service access cover
(4) Screwdriver

- Reset the alarm by pressing the **reset button** (→ 2/Fig. 52 Page 63)for 2 seconds.

Fig. 52: Aspiration smoke detector



(1) Status display
(2) Reset button
(3) Security tab
(4) Service access cover

» The aspiration smoke detector shows OK (→ 1/Fig. 52 Page 63).

- Start the system by pressing the **ON button** at the alarm box.

> If the reset fails, the concentration of smoke inside the detector chamber is too high. Re-try the reset after 1-2 minutes.

5.3 Gradient Supervision Espree

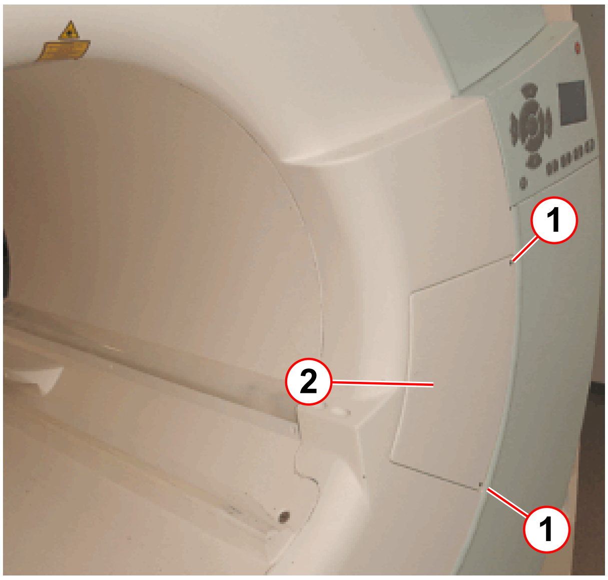
Required time: 40 min

Required tools: Smoke wick material number 10684293 or test spray material number 10684091, pocket lighter, screwdriver and 2.5 mm Allen key

Prerequisites: The display of the aspiration smoke detector shows OK (➔ 1/Fig. 63 Page 70). System is on.

- Remove the 2 Allen screws.
- Remove the service cover.

Fig. 53: Magnet service end



(1) Allen screws
(2) Service cover

⚠ WARNING

Accumulation of oxygen during helium filling.

The use of burning smoke wicks during helium filling could result in an open fire in the area of the magnet service turret.

» **Do NOT perform a smoke test during helium filling!**

SI Checking the gradient supervision

- Take the smoke wick out of the box.

Fig. 54: Smoke wick in the box

NOTICE

On-site smoke detectors

Avoid false alarm of the on-site smoke detector!

» **Check if there is an on-site smoke detector. The smoke or test gas must not enter the on-site smoke detector.**

» **Keep the smoke or test gas away from on-site smoke detectors.**

» **Keep the smoke time as short as possible to minimize smoke concentration in the air.**

- Light the tip of the smoke wick and allow it to burn for 10 - 15 seconds. Then blow out the flame.

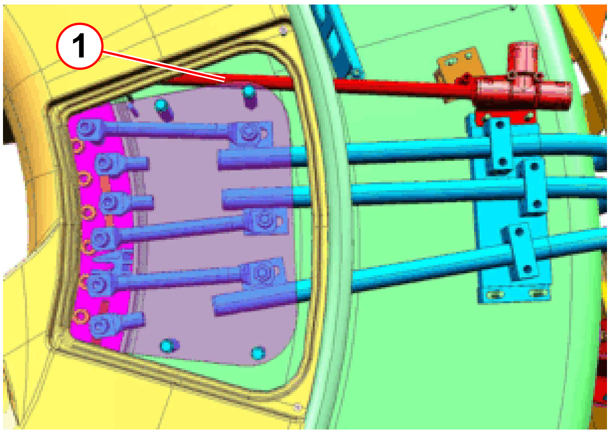
Fig. 55: Burning smoke wick

(1) Plastic handle

» The smoke wick provides a stable and continuous trail of smoke.

Fig. 56: Gradient coil connection / service access

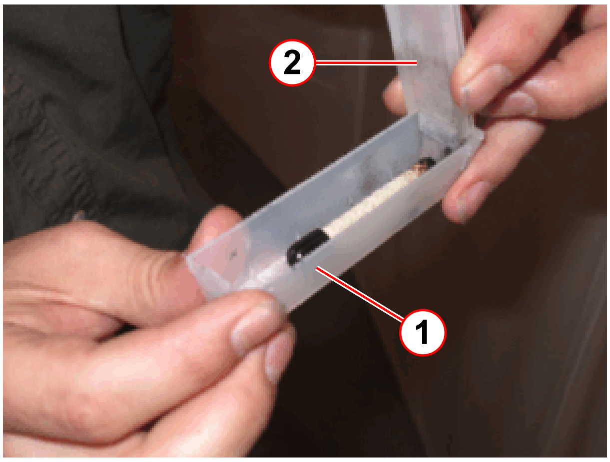
Fig. 57: Gradient coil connection / service access



(1) Intake manifold (The assembly of the manifold can vary)

- Hold the smoke wick behind the rear cover for 5 seconds so that the smoke is aspirated and transported to the detector (→ 1/Fig. 57 Page 66).
- After use, immediately place the smoke wick into the empty box and close the lid carefully.

Fig. 58: Extinguishing the smoke wicks



(1) Box
(2) Lid

» This will extinguish the smoke wick after approximately 15 seconds.

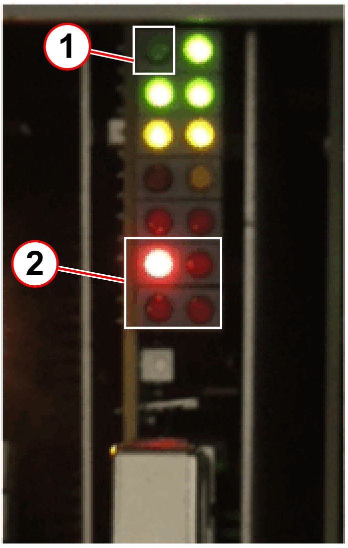
> ℹ Alternatively, the test spray can be used. Spray the test gas dosed toward the intake manifold for 5 seconds. Dosed means in this context to press the spray button not completely. Distance between spray and intake manifold should be approximately 15-20 cm.

Fig. 59: Test spray

Expected test results:

- The smoke is being sucked (driven by 2 small fans) from the gradient manifold to the aspirating smoke detector. This will take up to 40 seconds (depending on the hose length), then the following results are expected simultaneously:
  - » The red “FIRE” lamp on the VLM device must light up.
  - » The gradient modulator in the GPA / GSSU is disabled (red LED’s, error code 8).

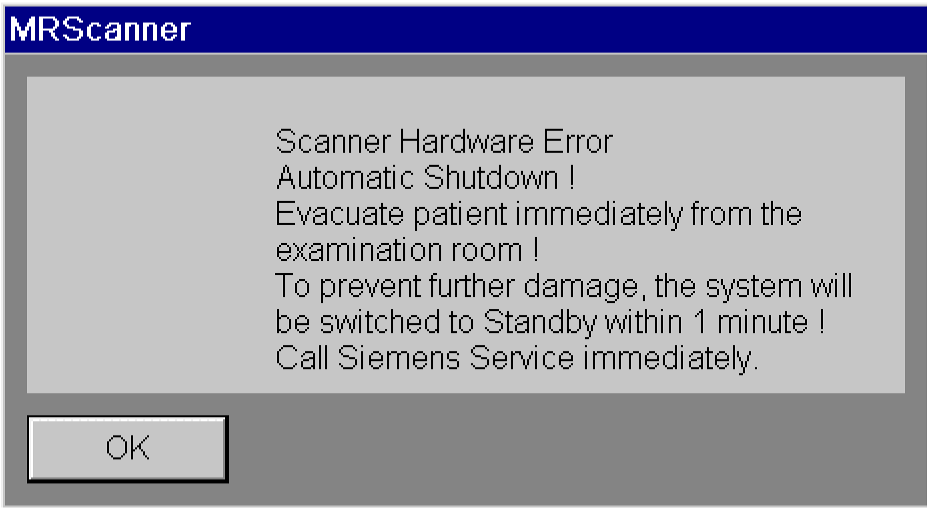
Fig. 60: Gradient modulator D40/ D41



(1) Modulator off
(2) Error code 8

» An alarm message appears at the main console. (can also be checked in the error log).

Fig. 61: Pop-up "Automatic Shutdown"



and the scanner icon shows error  .

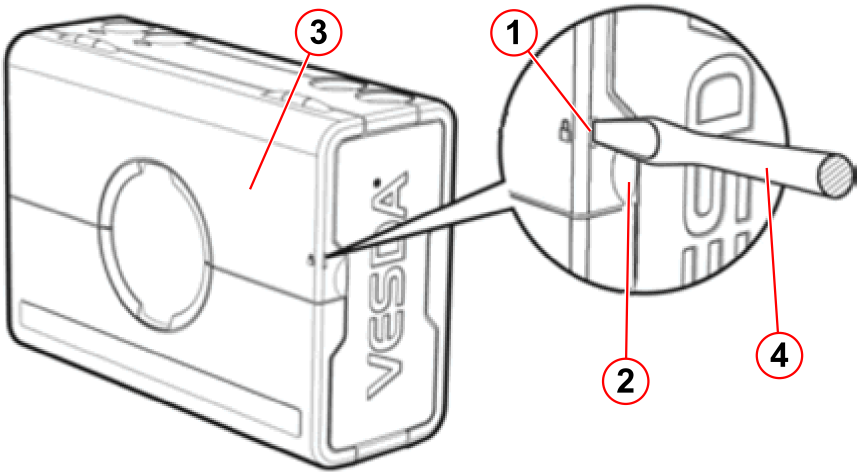
1 minute later,

» The system must be switched to stand-by mode automatically.

Reset the system:

- Press a screwdriver into the security tab (➔ 1/Fig. 62 Page 69) to open the service access cover with the finger clip (➔ 2/Fig. 62 Page 69).

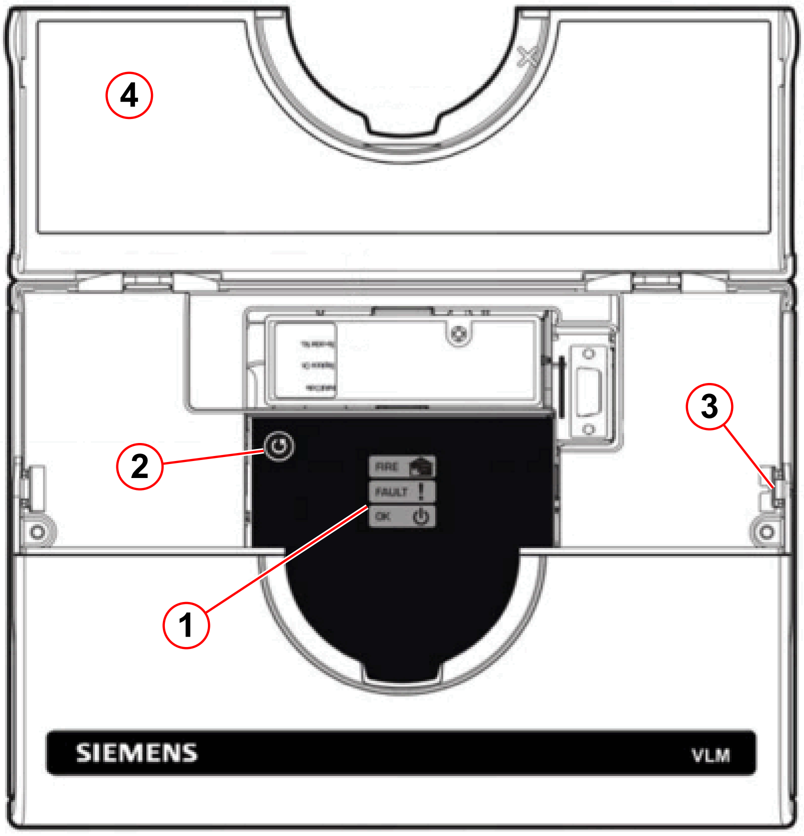
Fig. 62: Aspiration smoke detector ASD



(1) Security tab
(2) Finger clip
(3) Service access cover
(4) Screwdriver

- Reset the alarm by pressing the **reset button** (→ 2/Fig. 63 Page 70)for 2 seconds.

Fig. 63: Aspiration smoke detector



(1) Status display
(2) Reset button
(3) Security tab
(4) Service access cover

» The aspiration smoke detector shows OK (→ 1/Fig. 63 Page 70).

- Start the system by pressing the **ON button** at the alarm box.

> If the reset fails, the concentration of smoke inside the detector chamber is too high. Re-try the reset after 1-2 minutes.

5.4 Patient Table

5.4.1 Safety switches

Required time: 10 min

Required tools: Wooden block (delivered with the system beginning 2008)

If the wooden block is not on hand, one should be procured locally.

Recommended minimum size for softwoods: 89 mm x 89 mm; length approximately 540 mm.

SI Checking the safety switches

- Move the table all the way up.
- Position the wooden block under the support frame. Pay attention to the hole and to the hand crank on the underside of the support frame. The load on the wooden block and floor is approximately 200 kg. Place a board underneath the wooden block if the flooring is sensitive.

WARNING

Crushing hazard!

There is a potential crushing hazard when you lower the patient table.

» **Pay attention to your fingers when moving the patient table.**

Fig. 64: Example of timber block

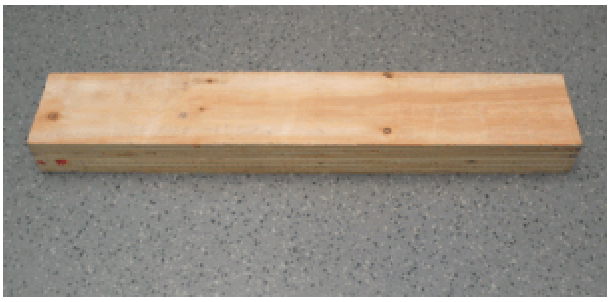
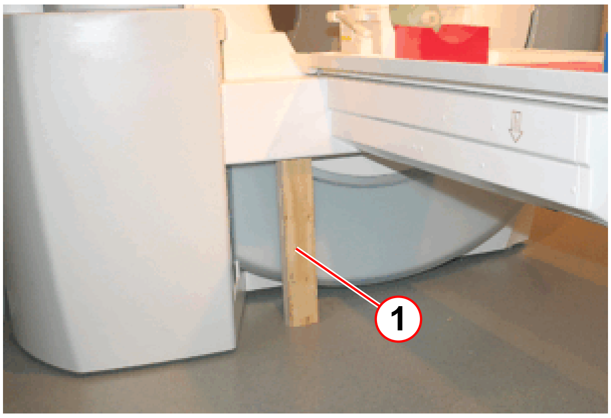


Fig. 65: Patient table / switch - S30 test



(1) Timber block

- Move the patient table carefully down until the support frame is resting on the wooden block and the spindle nut is mechanically disengaged.
  - » The spindle nut switch S30 opens and the magnet display shows ERROR 63.
  - » At the MRC, the pop-up window opens `< Patient Table Warning: Table Carrier Collision / Lift up table by using.......>`, press **OK**

Fig. 66: Magnet display

(1) ERROR (message) 63

» If ERROR 63 does not occur, troubleshooting is necessary.

- Move the table up by pressing the **Up/Inward key** on the control panel.
- Move the table all the way out by pressing the **Down/Outward key** on the control panel.
- Lift the patient table.

Fig. 67: "Lifting up" the patient table plate

Fig. 68: Magnet display

(1) Error (message) 64

» The magnet display shows `ERROR 64` while the tabletop plate is lifted .

» At the MRC, the pop-up window opens `< Patient Table Warning: ...>`, press **O.K.**

» If ERROR 64 does not occur, troubleshooting is necessary.

5.4.2 STOP function

Required time: 5 min

Required tools:n.a.

SI Checking the patient table stop function

- Activate and then deactivate the light marker, and then press the **center move** button. While the table is moving toward the isocenter, press the **STOP** button at the **left** side of the magnet aperture:

Verify the patient table stop procedure:

» The patient table stops; the display indicates the table stop (arrows blinking).

- Make sure that table movement is not possible.

» On the MRC, the pop-up window opens `< Patient table notice: STOP button pressed by user.........>`, press **OK**

- Deactivate the table **STOP** button by pressing the **IN** and **OUT** buttons in sequence.
- Press the center move button again to continue table movement. Move the table and press the **STOP** button on the **right** side of the magnet:
  - Repeat the steps above to verify the patient table stop procedure.
- Perform the function test for the **third control panel**, if present.

- Press the table **STOP button** on the **intercom at the MRC console**.
  - Perform to verify the patient table stop procedure.

5.4.3 Emergency movement

Required time: 5 min

Required tools:n.a.

SI Checking the patient table emergency movement

There are different types of drive mechanisms for the patient tables (exception: Trio A Tim System has only the whole body type).

The whole body drive has an emergency unlocking device. It is used to move the patient table manually.

The partial body patient drive requires no emergency unlocking device. It is possible to move the patient table manually.

Refer to the patient table (→ 1/Fig. 69 Page 75) to differentiate the table type.

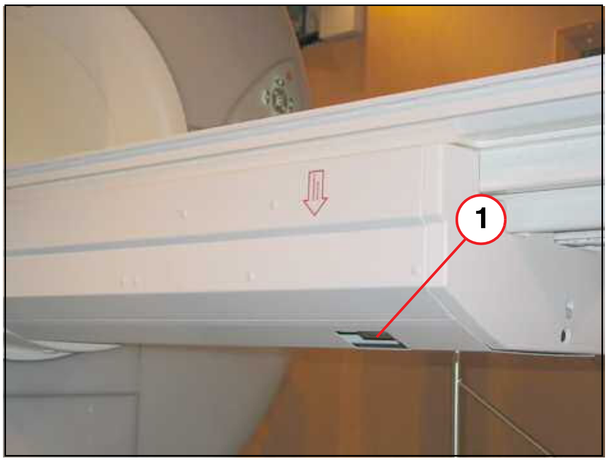
Partial body drive

- Move the table into center position by pressing the **center move** button.
- Press the patient table **STOP** button.
- Pull the table out of the magnet.

  » It must be possible to pull the table manually out of the magnet. The force required to move the tabletop plate depends on the speed. The slower you move, the less force you need.
- Deactivate the table **STOP** button by pressing the **IN** and **OUT** buttons in sequence.
- Confirm the output on the monitor.

Whole body drive

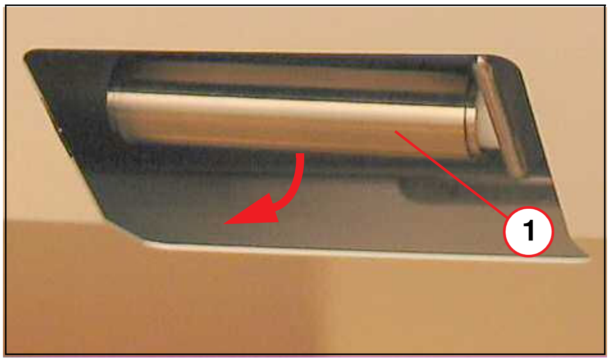
Fig. 69: Patient table



(1) Emergency release

- Move the table into center position.

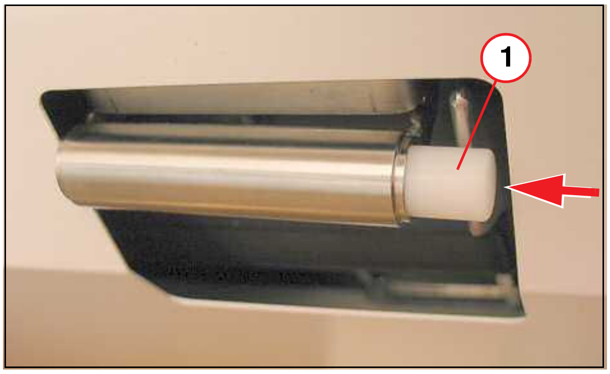
Fig. 70: Emergency release



(1) Handle

- Pull the emergency release handle on the patient table.
  » It must be possible to pull the table manually out of the magnet.

Fig. 71: Emergency release



(1) Release button

- Press the release button (1) on the emergency release.
- Move the table manually, approximately 10 cm, until the table locks into place. And the handle returns to its original position.
- Confirm the output on the monitor.
- To reset the table, alternately press the IN and OUT buttons on the scanner control panel.
- Press the home button to move the table back into the home position.

5.4.4 Patient table bruising hazard

5.4.4.1 Service end

SI Checking the distance between the patient table and the wall in the RF-room

Required time: 5 min

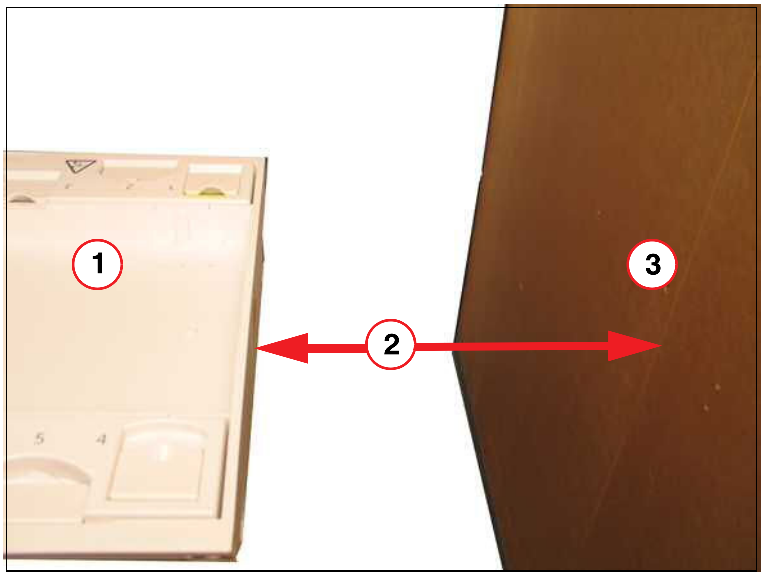
Required tools: bruising hazard sign p/n 3861718, tape p/n 3474913; if required

- Pull the emergency release handle on the patient table (whole body type) or press the patient table **Stop** button (partial body type).
- Manually move the table all the way back (toward the magnet service end) until the table stops in the horizontal end position.

  » If the distance (→ 2/Fig. 72 Page 76) between the patient table and the wall (service end) of the RF-room is **greater than** 500 mm, no further actions are necessary. Perform final steps.

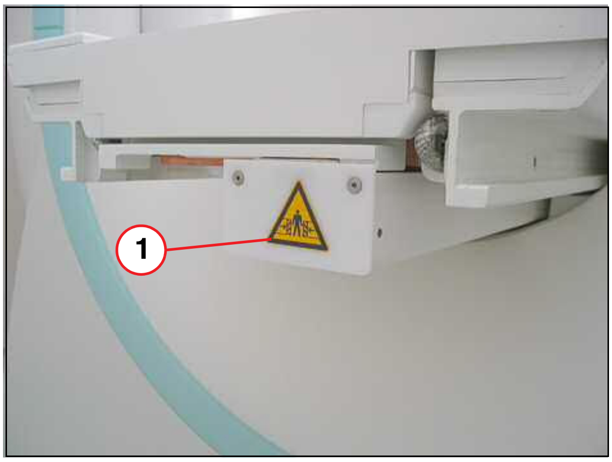
  » If the distance (→ 2/Fig. 72 Page 76) between the patient table and the wall (service end) of the RF-room is **less than** 500 mm, the bruising hazard sign (→ 1/Fig. 73 Page 77)has been posted in back of the patient table and the controlled access area has been protected with a yellow-and-black tape (→ 1/Fig. 75 Page 77).

Fig. 72: Bruising hazard - controlled access area



(1) Patient table back in horizontal end position
(2) Distance
(3) Wall of the RF room

Fig. 73: Bruising hazard

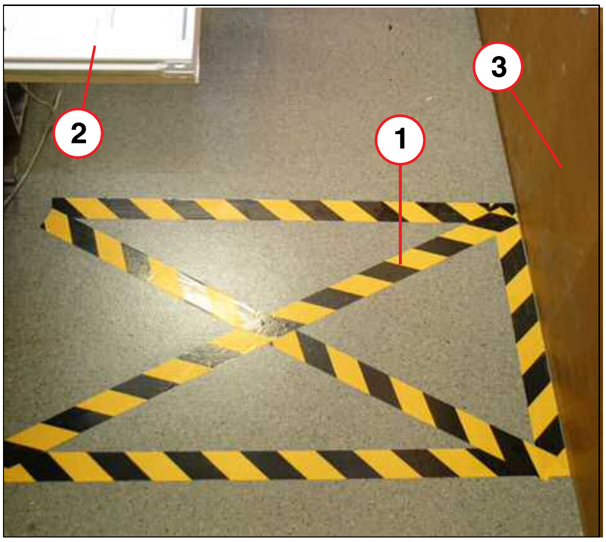


(1) Label

Fig. 74: Bruising hazard sign

- Ensure that the bruising hazard sign (→ 1/Fig. 73 Page 77)has been posted in back of the patient table if the distance between patient table and wall (service end) is less than 500 mm.
- Install missing sign immediately

Fig. 75: Service end area



(1) Marking on the floor
(2) Patient table
(3) Wall of the RF room

- Ensure that the controlled access area has been protected with a yellow-and-black tape, if the distance between patient table and wall (service end) is less than 500 mm.
- Install missing tape immediately.

Final steps:

1. Slide the table back into the magnet.
2. Press the release button on the emergency release (whole body type).
3. Move the table manually, approximately 10 cm, until the table locks into place and the handle returns to its original position(whole body type).
4. Confirm the output on monitor.
5. To reset the table, alternately press the IN and OUT buttons on the control panel of the scanner.
6. Press the home button to move the table back into the home position.

5.4.4.2 Cover

SI Checking the distance between the patient table and the cover

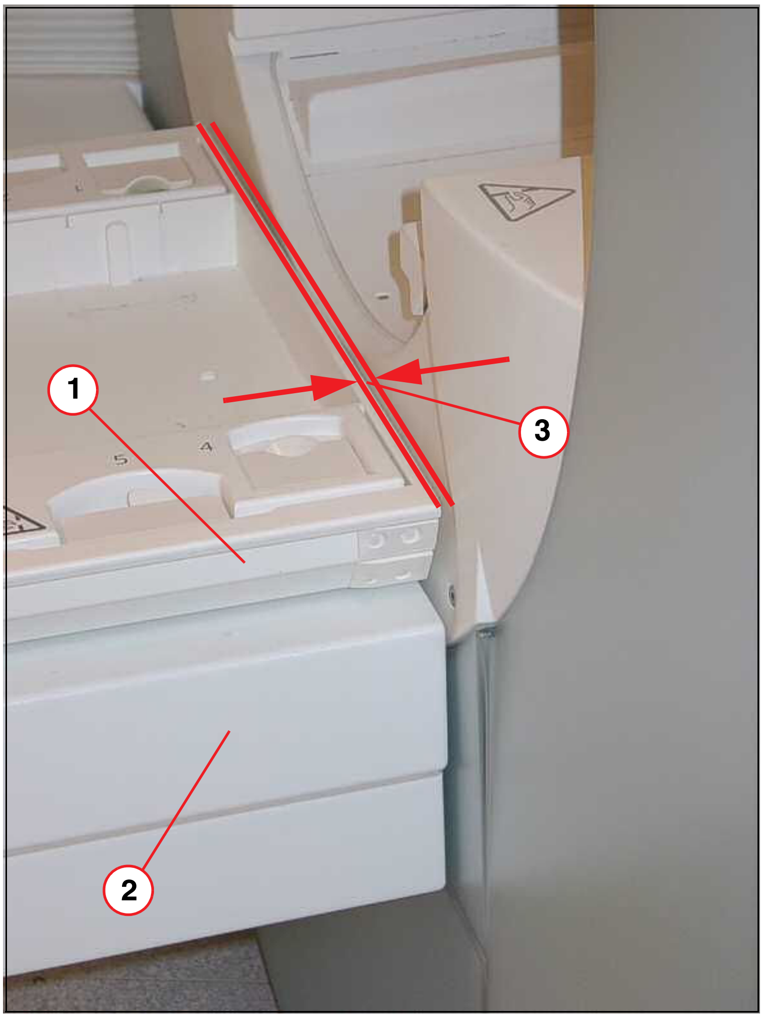
Required time: 5 min

Required tools:n.a.

> The translucent front cover may not be exactly vertical. As a result, the translucent front cover and table may not be in parallel. In this case, do not try to compensate for the angle between the table and the cover using the X-radial adjustment procedure. The procedure also changes the travel path of the tabletop plate inside the magnet bore.

> The following pictures provide general information on how to perform the distance checks. The cover design differs slightly among Tim systems.

Fig. 76: Clearance, vertical gap



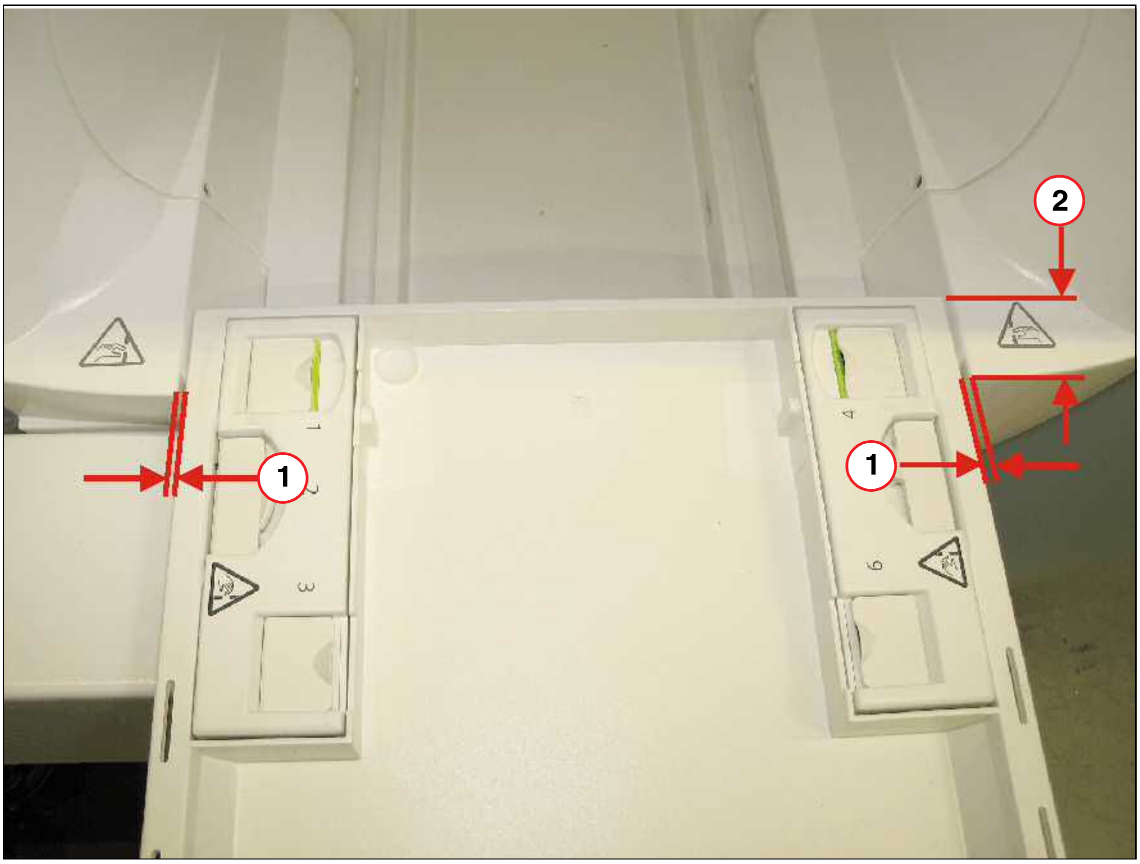
(1) Tabletop
(2) Support frame
(3) Clearance < 8 mm between table and magnet cover

1. Move the patient table up to the highest position and then lower it about 80 mm.
2. Measure the distance of the vertical clearance (➔ 3/Fig. 76 Page 79) between the tabletop plate and the magnet cover.
   » The distance between the tabletop plate and the magnet cover (➔ 3/Fig. 76 Page 79) must be within the range of **< 8 mm**.
   » If the distance is not within range, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

The patient table shall run in parallel to the body coil and covers with a sufficient clearance across the entire tabletop plate to prevent scratching.

There is always clearance between the replaceable tabletop plate and the table carrier, and between the table carrier and the slide way. For this reason, the replaceable tabletop plate must be pushed as described.

Fig. 77: Case 1: Checking the distance between patient table and cover (patient end)



(1) Distance to be measured
(2) Approx. 50 mm

Case 1:

1. Move the patient table approximately 50 mm (for Symphony, A Tim System 400 mm) into the magnet.
2. Push the tabletop plate (replaceable tabletop plate) to the **left** side.

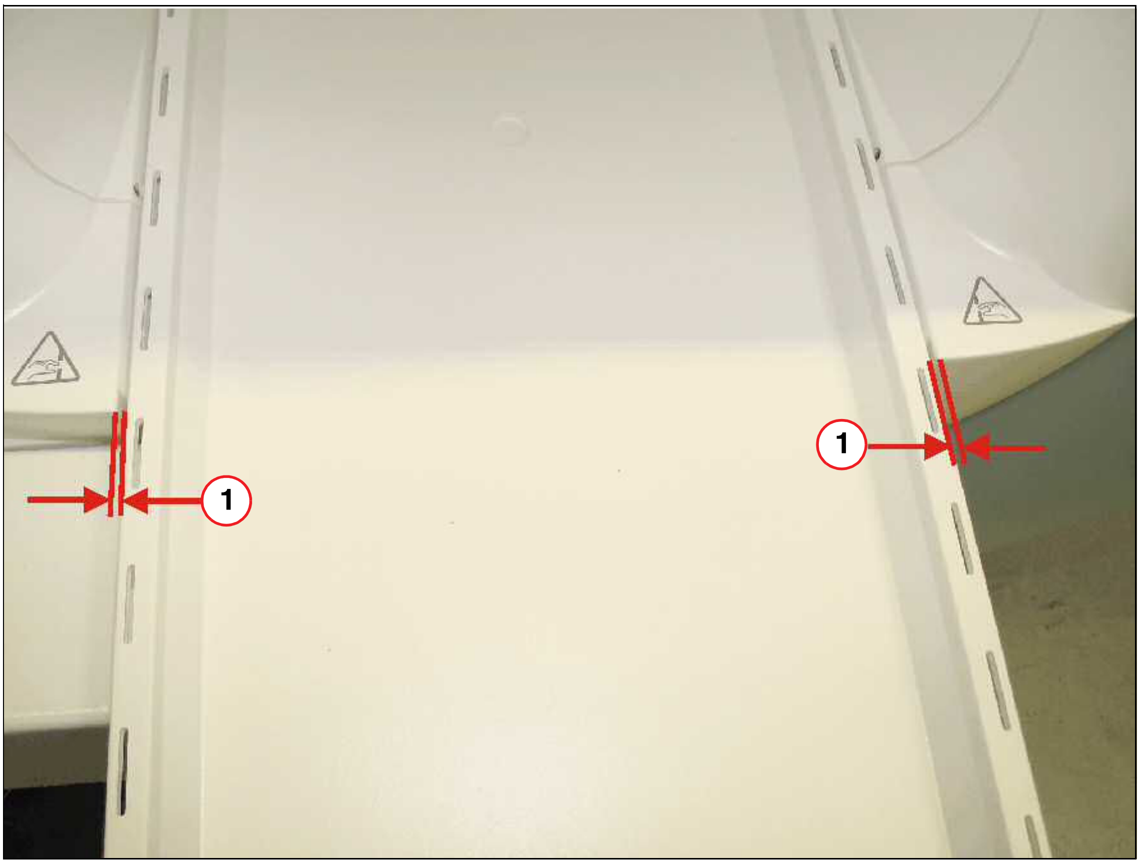
   Check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.
3. Repeat the procedure by pushing the tabletop plate to the **right** side

   Check the distance of < 8 mm on the **left** side, at the front of the magnet.

   » The distance in both tests has to be within the range of **< 8 mm**.

   » If the distance is not within the range of **< 8 mm**, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

Fig. 78: Case 2: Checking the distance between patient table and cover / body coil (patient end)



(1) Distance to be measured

Case 2:

1. Move the patient table approximately 1200 mm into the magnet.
2. Push the tabletop plate (replaceable tabletop plate) to the **left** side.

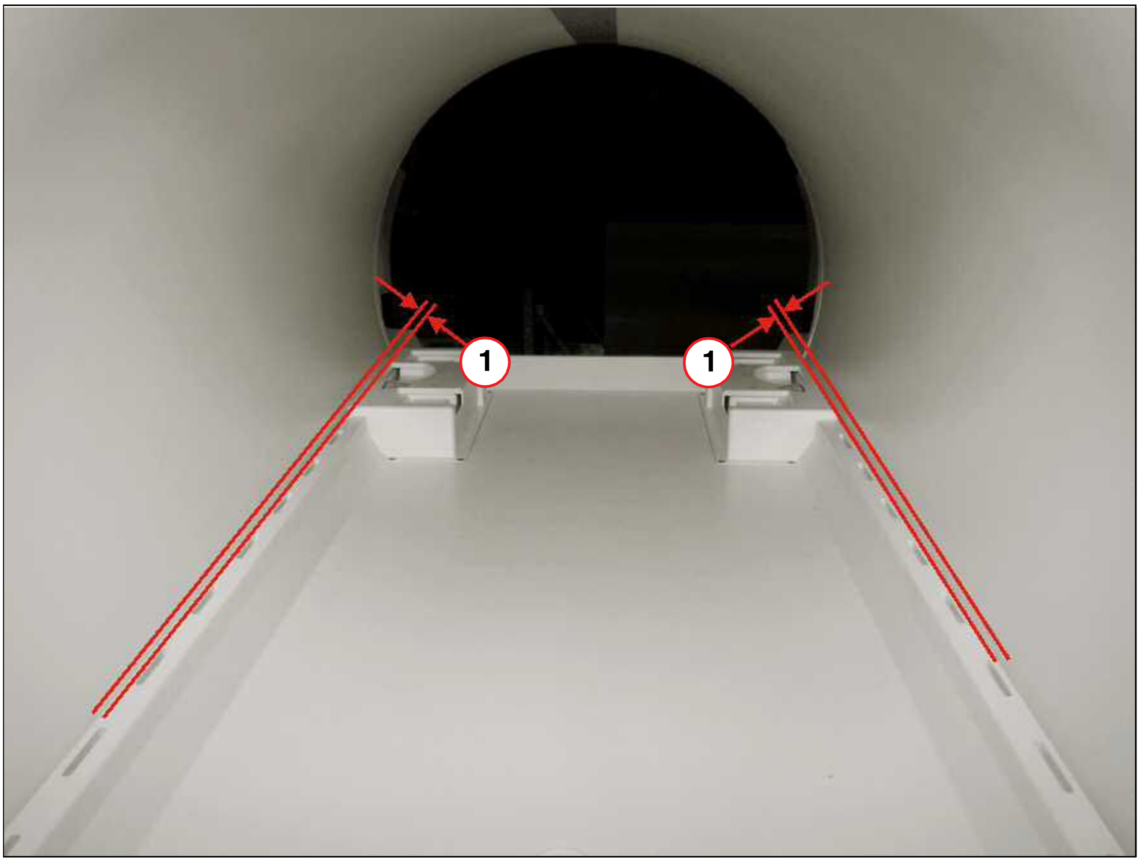
   Check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.
3. Repeat the procedure by pushing the tabletop plate to the **right** side.

   Check the distance of < 8 mm on the **left** side, at the front of the magnet.

   » The distance in both tests has to be within the range of **< 8 mm**.

   » If the distance is not within the range of **< 8 mm**, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

Fig. 79: Case 3: Checking the distance between patient table and Body Coil / cover (service end)



(1) Distance to be measured

Case 3:

1. Move the complete patient table into the magnet.
2. Check the distance of < 8 mm between the patient table and cover / body coil on both sides along the entire length of the tabletop plate.
   » The distance has to be within the range of **< 8 mm**.
   » If the distance is not within the range of **< 8 mm**, the patient table must be adjusted. Refer to Installation > Start-up > Ramping the magnet > Check the mechanical adjustment of the patient table.

5.4.5 Squeeze bulb

Required time: 5 min

Required tools:n.a.

SI Checking the squeeze bulb

Note: Perform the following test during an MR measurement.

The door to the RF-room shall be closed (direct acoustic coupling between the RF-room and the operating console is momentarily down).

- Press the squeeze bulb at the patient table.

The following must occur:

- » A continuous audible signal is heard at the intercom on the operating console (MRC).
- » The red LED above the speak button is flashing.
- » The audible signal and the red LED go off when you press the “speak” button.
- » Communication with the patient is established when you press the “listen” button on the control unit (MRC).
- » You can speak with the patient when you press the “speak” button on the operating console.
- » You change the volume of the intercom by turning the “volume1” (speaker, intercom) and “volume 2” (speaker, cabin) button as required.

5.4.6 Table movement

Required time: 5 min

Required tools:n.a.

SI Checking the patient table movement

Prerequisites: The system is running.

- Move the table all the way out.
  - » The table must move without any noticeable noise.
  - » The table stops in the horizontal end position.
- Lower the table all the way.
  - » The table must move without any noticeable noise.
  - » The table stops in the vertical end position.
- Move the table all the way up again.
  - » The table must move without any noticeable noise.
  - » The table stops in the vertical end position.
- Move the table all the way in.
  - » The table must move without any noticeable noise.
  - » The table stops in the horizontal end position.
- Move the table to the Home position.

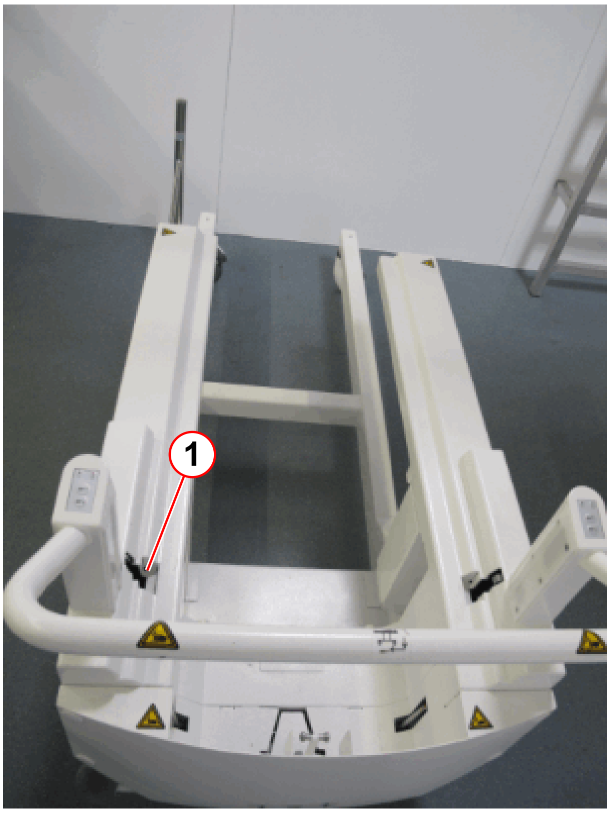
5.5 Patient Trolley

5.5.1 Option: Trolley

Required time: 10 min

Required tools: n.a.

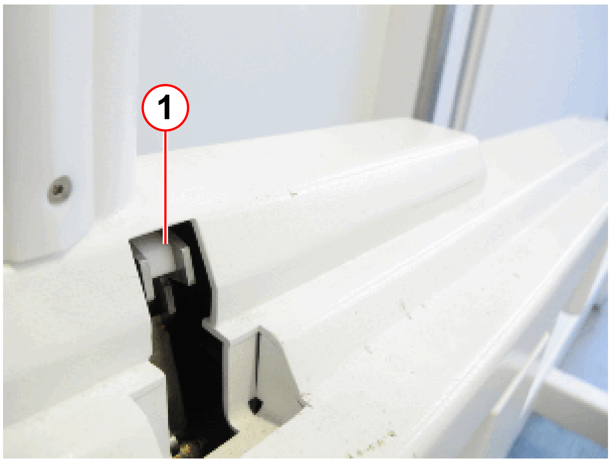
Fig. 80: Trolley



(1) Rastklinke

Trolley docking mechanism.
(➔ 1/Fig. 80 Page 84)

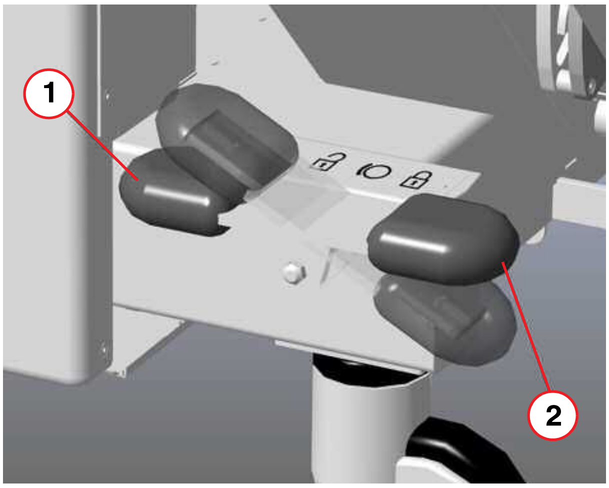
Fig. 81: Engaging pawl



(1) Plugged-in position

- Check the “plugged- in” position of the left engaging pawl.
(➔ 1/Fig. 81 Page 84)

Fig. 82: Trolley brake, foot end



(1) Pedal side: Release
(2) Pedal side: Locking mechanism

- Press down on the brake pedal (2)
  🔒

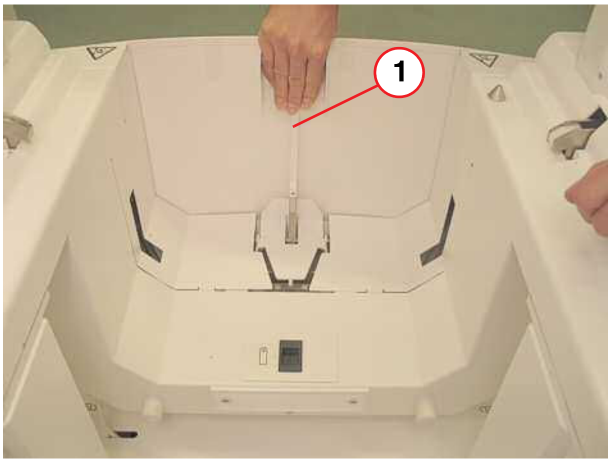
> ℹ If the removable tabletop is not locked, the transfer configuration of the trolley is always in its lowest position. The height of the transfer configuration can be adjusted only after transfer of the removable tabletop to the trolley.

Condition: Trolley without removable tabletop.

- Try to push down the lifting pedal.
  - » It must not be possible to push down the lifting pedal.

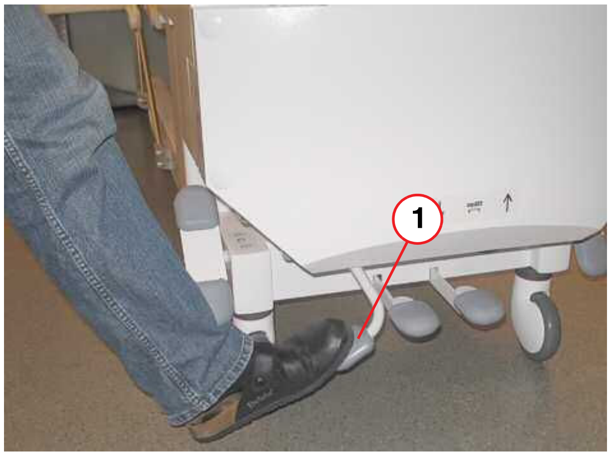
1. Push the lever backward and hold the lever in this position (➔ 1/Fig. 83 Page 85).

Fig. 83: Trolley



(1) Lever

Fig. 84: Trolley locking/releasing pedal



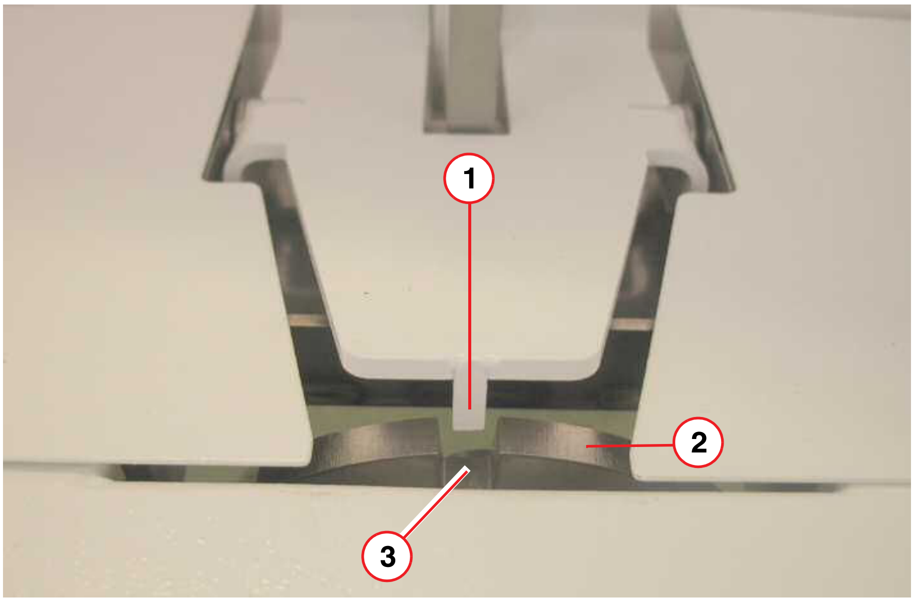
2. Step down fully on the locking/releasing pedal (➔ 1/Fig. 84 Page 85).
   - » The locking/releasing pedal locks audibly into position and remains in the inclined position.

3. Check the position of the disk (➔ 2/Fig. 85 Page 86).
   » The slot (➔ 3/Fig. 85 Page 86) has to be positioned aligned below the nib (➔ 1/Fig. 85 Page 86).

   Slowly move the lever (➔ 1/Fig. 83 Page 85) forward. The nib must not grind to a slot side.

   In case of misalignment, repair in the factory is required.

Fig. 85: Trolley



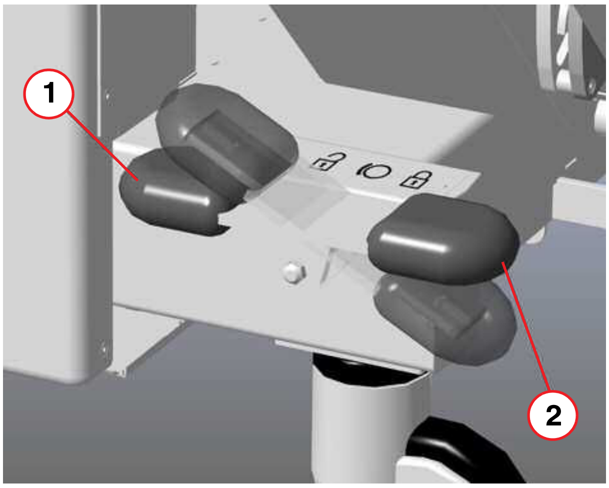
(1) Nib
(2) Disk
(3) Slot

4. Push the lever backward and hold the lever in this position (➔ 1/Fig. 83 Page 85).
5. Step fully on the locking/releasing pedal (➔ 1/Fig. 84 Page 85) and let it swing back into the rest position.
6. Release the trolley brakes.

Fig. 86: Patient table, load simulation

1. Move the tabletop plate into the home position.
2. Position several coils and phantoms on the tabletop plate (load simulation).
3. Disconnect all coil plugs and close all protective covers at the coil plug-in locations.
4. Disconnect all hoses (squeeze bulb, vacuum cushion, headphones).
5. Dock the trolley to the patient table.

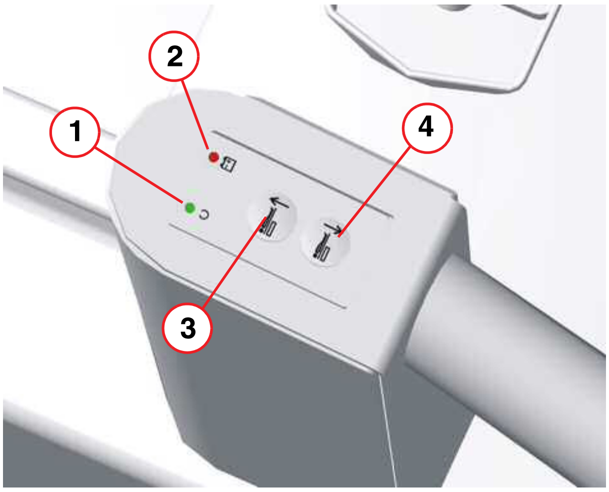
Fig. 87: Trolley brake, foot end



(1) Pedal side: Release
(2) Pedal side: Locking mechanism

- Press down on the brake pedal (2)

Fig. 88: Trolley remote control



(1) Operation LED
(2) Battery LED
(3) Key for raising the removable tabletop
(4) Key for lowering the removable tabletop

- Press the **Lower key** (4) on the remote control to lower the tabletop plate until it stops by itself. While doing this, check both remote control keys to the right and left of the trolley.

- Step down fully on the locking/releasing pedal.(→ 1/Fig. 84 Page 85)

Fig. 89: Trolley locking/releasing pedal

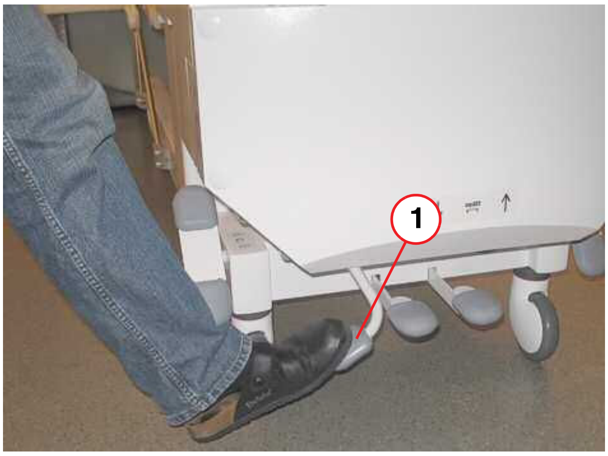
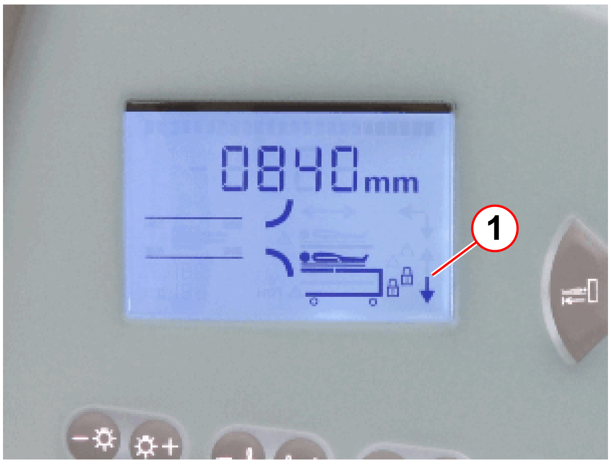


Fig. 90: Display, docked trolley



(1) Down arrow

» The locking/releasing pedal locks audibly into position and remains in the inclined position.

» The arrow down on the display must appear, when the pedal was pushed completely (→ 1/Fig. 90 Page 88)

» If the arrow is visible before the locking pedal is locked completely the engaging pawl must be adjusted.

Refer to: **Replacement of Parts -> Patient Handling-> Locking Hooks**

» The removable tabletop is secured at the trolley.

» The table display shows the request to disconnect the connectors for the coils as well as other components (hoses for squeeze bulb etc.)

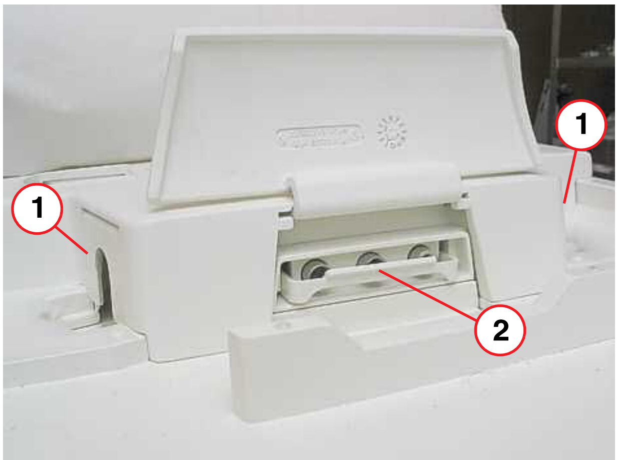
NOTICE

Damage of coils and tabletop plate

When lowering the carrier plate while the coil plugs or hoses are connected, both the removable tabletop and the coils may sustain considerable damage.

» **Ensure that all hoses and coils connectors have been disconnected and all protective covers at the coil plug-in locations are closed (→ Fig. 91 Page 89).**

Fig. 91: Patient table / foot end



(1) Protective covers / coil plug-in
(2) Protection bracket

1. Press the **Lower key** at the remote control and keep it pressed until the carrier plate has been lowered completely.
2. Release the trolley brake (→ 1/Fig. 87 Page 87).
3. Move the trolley including the tabletop plate away from the patient table.

SI Option trolley: Checking the locking mechanism

Fig. 92: Trolley: Head end (magnet side)

- Activate all trolley brakes.
- Shake the head end of the removable tabletop. You are checking the locking mechanism.

- Release all trolley brakes and move the trolley back to the patient table (→ 1/Fig. 87 Page 87).

- Dock the trolley on the patient table.
- Activate the trolley brakes(→ 1/Fig. 87 Page 87) .
- Press the **Raise key** (→ 3/Fig. 88 Page 88) on the remote control to lift the table carrier plate until it stops by itself. While doing this, check both remote control keys to the right and left of the trolley.
- Step fully on the locking/releasing pedal (→ 1/Fig. 87 Page 87) and let it swing back into the rest position.
- Press the **Raise key** on the remote control
- Release the trolley brakes (→ 1/Fig. 87 Page 87).
- Move the trolley away from the patient table.

5.6 ERDU

There are two types of magnet supervisions (MSUP K2256, MSUP K2306). The test procedures are different. Check the ERDU function according to the MSUP type.

Use the procedure (➔ ERDU (MSUP K2306) / Page 102) for MSUP K2306.

Use the procedure (➔ ERDU / Page 91) for MSUP K2256.

5.6.1 ERDU

Required time: 30 min.

Required tools:

ERDU Test Load III (ETL) - 03861197

Adaptor cable B - 10129824 (if required, this adaptor should be in the ETL kit),

MSUP Display-Unit - 07758522,

Flat bladed screwdriver (non-magnetic).

The Adaptor cable B is available from CSML as a spare (part no. 10129824)

The Mains cable supplied with the ETL is a UK, Euro and a US AC cable. When the ETL is used in another country an IEC 320 C13 mains lead for that country must be obtained.

SI ERDU Test

5.6.1.1 Prerequisites

WARNING

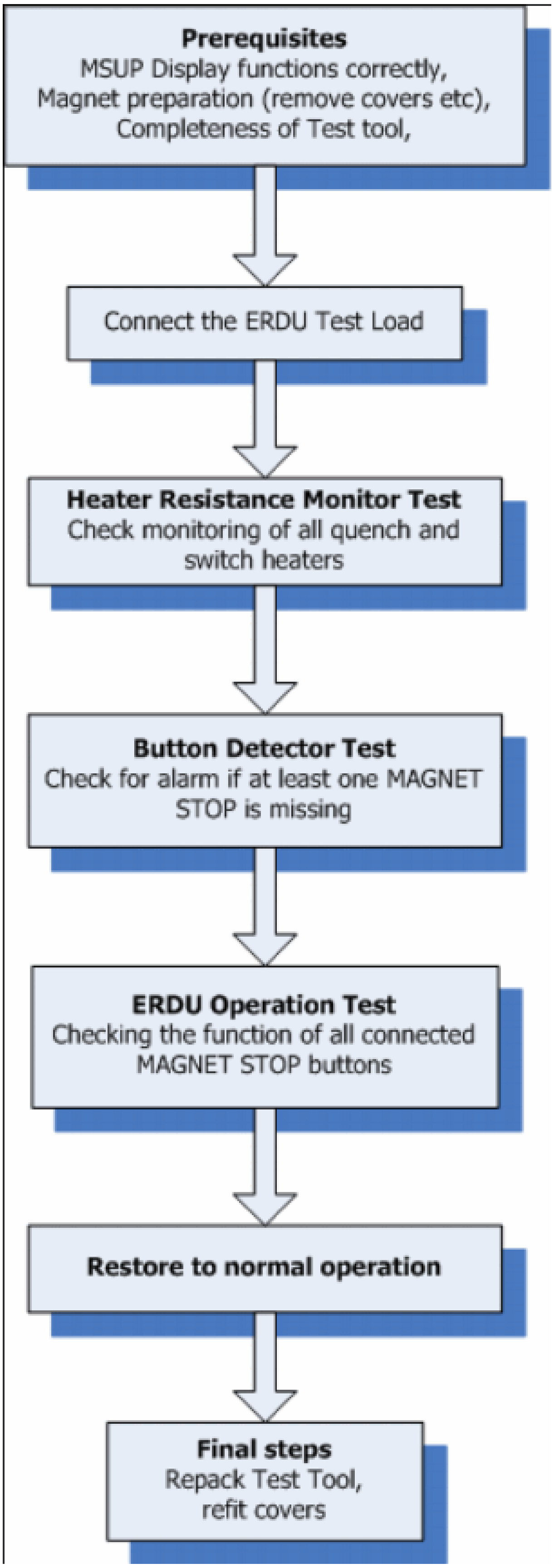
Restrict Access.

Danger of Ferrous materials entering imaging suite.

Unauthorized personnel must be excluded from the imaging suite.

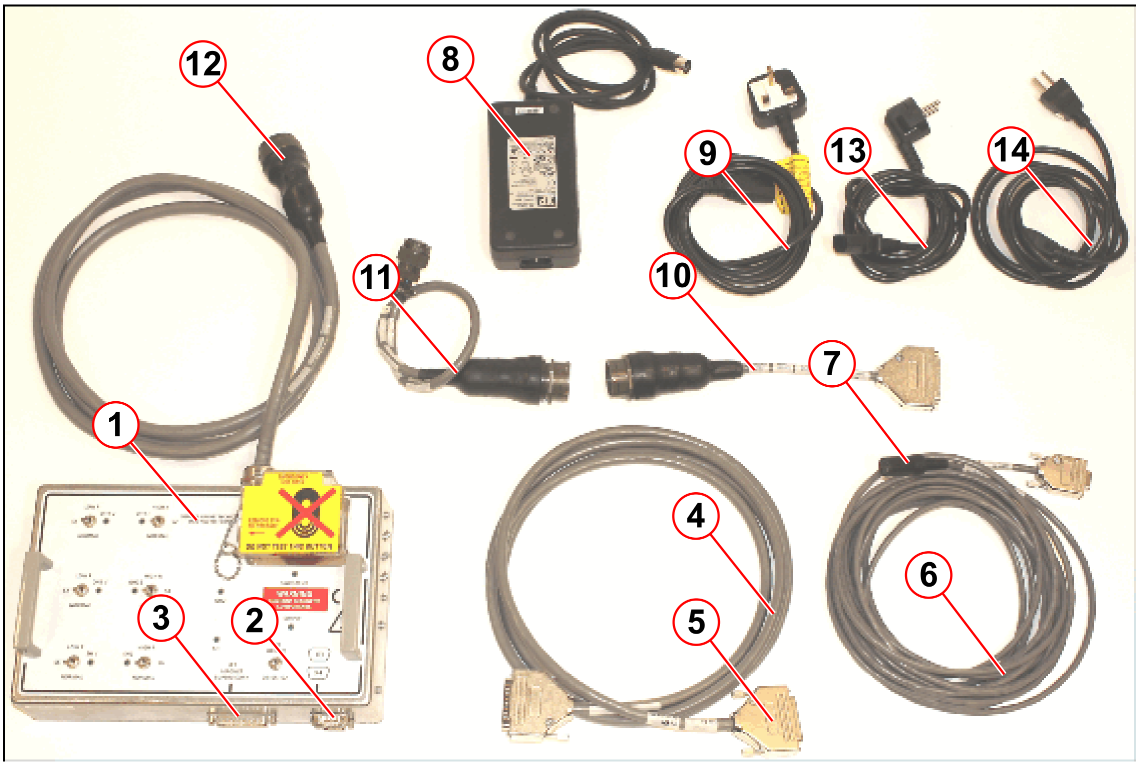
» **Secure the room and prevent unauthorized access.**

Fig. 93: ERDU testing flow chart



- Restrict access to the magnet if it is at field while the ETL is connected to the MSUP.
- Check that all alarm and fault LED’s on the alarm box show that the system is OK.
- Check that the **ERDU Test Load Kit** is complete with all the listed parts present:

Fig. 94: ERDU Test Load Kit



(1) ERDU Test Load III (ETL)
(2) 9-pin Connection X3 or X4 for PSU
(3) 25-pin Connection X1
(4) MSUP - ETL Cable
(5) 25-pin Connector to MSUP X1
(6) PSU Output Extension Lead
(7) PSU Output Extension Lead Connector to PSU
(8) Power Supply Unit (ETL PSU)
(9) AC Supply Lead (UK)
(10) Adaptor Cable B
(11) Adaptor Cable for OR64 / OR70, 41-pin Connector to 10-pin Magnet Turret Connector 'B'
(12) 41-pin Connector, Magnet Turret
(13) AC Supply Lead (EURO)
(14) AC Supply Lead (US)

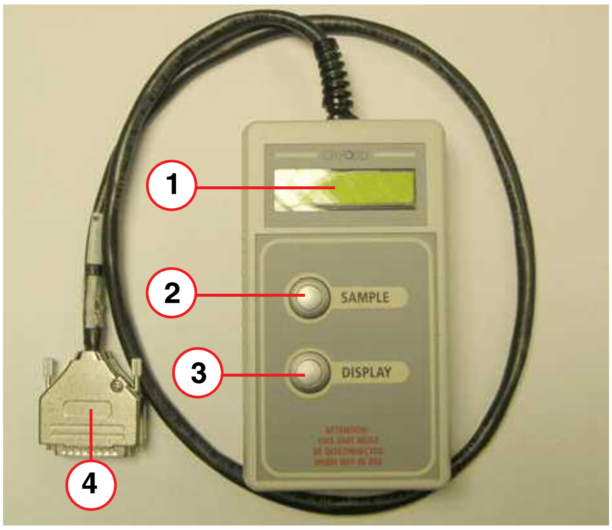
- Remove the side cover:
  - Right side cover (early Avanto systems).
  - Left service access cover (Espree, Avanto, Verio).
- Connect the **MSUP-Display Unit** to **X9** on the MSUP.

Helium remote display unit.

> The helium remote display unit is a Service Tool and must be disconnected during imaging to avoid interference to the MR Signal.

General

Fig. 95: MSUP- Display Unit



(1) LCD Display
(2) Sample button
(3) Display scroll button
(4) 25 way 'D' connector

The helium remote display unit is a monitoring device that is connected to **X9** of the MSUP. It provides local monitoring of the magnet parameters, including alarms and warnings if values are not within specification. In addition, some functions can be toggled **ON** or **OFF** manually.

The unit will also operate in battery mode from the MSUP battery, to provide liquid helium measurement during transportation/installation.

Connecting the Helium remote display unit

- Connect the Unit to X9 of the MSUP.
- Press both the **SAMPLE** (2) and **DISPLAY** buttons (3), at the same time, for approximately **5 seconds.**
- Press the **SAMPLE** to start the test.
  - The display will show **'TEST IN PROGRESS'** whilst carrying out an internal **Self Test** and a helium level sample.

Display Magnet Parameters

- Press the **DISPLAY** button repeatedly to display the different monitored sections. The sections used during the ERDU test are 7, 8 and 10.

If a parameter is outside it's specified limits, a "W" or "A" will be displayed next to the parameter reading to indicate a **WARNING** or **ALARM**. If this is the situation then trouble shoot the error.

Tab. 8 OR105/122 MSUP-Display Parameters

Item #	Parameter	Typical Value +/-20% Tolerance
7	Switch Heater (Ohms, status)	130 Ohms, Off
8	Quench heaters 1, 2 (Ohms)	22.5 Ohms, 22.5 Ohms

5.6.1.2 Connecting the ERDU Test Load (ETL)

WARNING

It is not possible to quench the magnet using the normal system MAGNET STOP buttons while the ERDU Test Load is connected to the MSUP and magnet.

» **The magnet may be quenched only by using the MAGNET STOP button on the ERDU Test Load.**

» **Restrict access to the magnet accordingly.**

- The **ETL PSU** has to stay outside the magnet room. Connect it to a convenient AC outlet using the mains lead for your country.

Ensure that the ETL PSU is as far away from the magnet as possible (magnetic field < 50 Gauss).

- Connect the ETL PSU to the ETL **X3** or **X4** using the PSU output extension lead (➔ 2/Fig. 94 Page 93)

  » Check that the **SUPPLY** and **SWITCH OK** indicators glow on the ETL.

WARNING

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

» **Ensure that the MAGNET STOP button on the ETL is NOT OPERATED!**

» **DO NOT operate the MAGNET STOP button on the ERDU Test Load unless an emergency magnet quench is required.**

- Connect the **MSUP - ETL** cable to the 25-pin connector **X1** on the ERDU Test Load. (➔ 3/Fig. 94 Page 93)
- Set SH for correct magnet type (OR64/70 or OR105/122) .
  - For ETL 2660, the SH switch is not available. The service tool is only configured for OR105/122.
- Set all **HIGH R / LOW R** switches on the ETL to **NORMAL** position (downward).
- Connect the **adaptor cable B** to the **41-pin turret connector cable**. (➔ 12/Fig. 94 Page 93)

- Disconnect the lead **X1** on the MSUP and connect the end of **adaptor cable B** (➔ 10/Fig. 94 Page 93)onto the cable which was plugged into **X1**.
- Check that the alarm sounds and that the **MAG STOP** supply indicator lights up on the **Alarm Box**. (➔ Fig. 97 Page 97).
  » Press **ACKNOWLEDGE** to cancel the audible alarm.
- Check that the **QH1, QH2, SH** and **SWITCH OK** and **SUPPLY** indicators glow on the ETL.
- Connect the **ETL- MSUP lead** (➔ 5/Fig. 94 Page 93) to **X1** on the MSUP.

> The ETL will sound a continuous buzzing alarm, until the ETL is disconnected.

  » Check **audible alarm** sounds in the ETL.
- Silence any alarms on the Alarm Box by pressing the **ACKNOWLEDGE** button, if re-quired.

5.6.1.3 Heater resistance monitor tests

Resistance figures are nominal- +/-20% tolerance are acceptable.

Fig. 96: ERDU Test Load (front panel)

Fig. 97: Alarm Box (buttons and controls)

1. Operate the **SH LOW R** switch on the ERDU Test Load.
   » Check the Alarm Box buzzer is ON and MAG STOP LED ON.
   » Press the **ACKNOWLEDGE** button at the Alarm Box. Check the Alarm box buzzer turns off.
   » Check for MSUP-Display Unit indication 7. SWITCH HEATER **50R A.**
2. Reset the SH LOW R switch on the ERDU Test Load to normal (downwards).
   » Check for MSUP-Display Unit indication 7. SWITCH HEATER **130R.**

1. Operate the **SH HIGH R** switch on the ERDU Test Load.
   » Check the Alarm Box buzzer is ON and MAG STOP LED ON.
   » Press the **ACKNOWLEDGE** button at the Alarm Box. Check the Alarm box buzzer turns off.
   » Check for MSUP-Display Unit indication 7. SWITCH HEATER **250R A.**
2. Reset the SH HIGH R switch on the ERDU Test Load to normal (downwards).
   » Check for MSUP-Display Unit indication 7. SWITCH HEATER **130R.**

1. Operate the **QH1 LOW R** switch on the ERDU Test Load.
   » Check for Alarm Box buzzer ON and MAG STOP LED ON.
   » Press the **ACKNOWLEDGE** button at the Alarm Box. Check the Alarm box buzzer turns off.
   » Check for MSUP-Display Unit indication 8. QUENHTR **8R A 21R**.
2. Reset the QH1 LOW R switch on the ERDU Test Load to normal (downwards).
   » Check for MSUP-Display Unit indication 8. QUENHTR **21R 21R**.

1. Operate the **QH1 HIGH R** switch on the ERDU Test Load.
   » Check for the Alarm Box buzzer ON and MAG STOP LED ON.
   » Press the **ACKNOWLEDGE** button at the Alarm Box. Check the Alarm box buzzer turns off.
   » Check for MSUP-Display Unit indication 8. QUENHTR **42R A 21R**.
2. Reset the QH1 HIGH R switch on the ERDU Test Load to normal (downwards).
   » Check for MSUP-Display Unit indication 8. QUENHTR **21R 21R**.

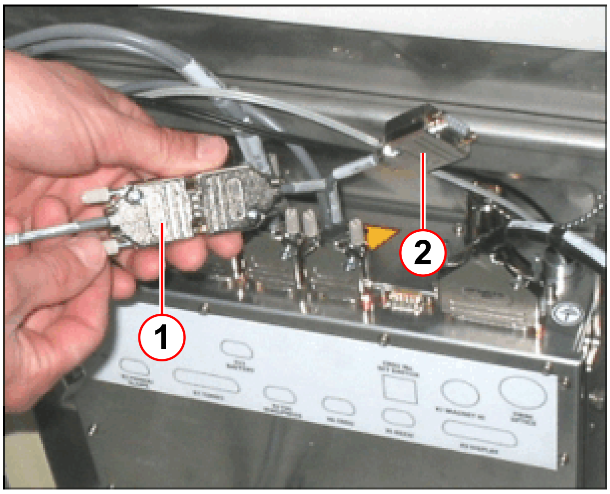
1. Operate the **QH2 LOW R** switch on the ERDU Test Load.
   » Check for Alarm Box buzzer ON and MAG STOP LED ON.
   » Press the **ACKNOWLEDGE** button at the Alarm Box. Check the Alarm box buzzer turns off.
   » Check for MSUP-Display Unit indication 8. QUENHTR **21R A 8R**.
2. Reset the QH2 LOW R switch on the ERDU Test Load to normal (downwards).
   » Check for MSUP-Display Unit indication 8. QUENHTR **21R 21R**.

1. Operate the **QH2 HIGH R** switch on the ERDU Test Load.
   » Check for Alarm Box buzzer ON and MAG STOP LED ON.
   » Press the **ACKNOWLEDGE** button at the Alarm Box. Check the Alarm box buzzer turns off.
   » Check for MSUP-Display Unit indication 8. QUENHTR **21R A 42R**.
2. Reset the QH2 HIGH R switch on the ERDU Test Loadto normal (downwards),.
   » Check for MSUP-Display Unit indication 8. QUENHTR **21R 21R**.

5.6.1.4 Button detector test

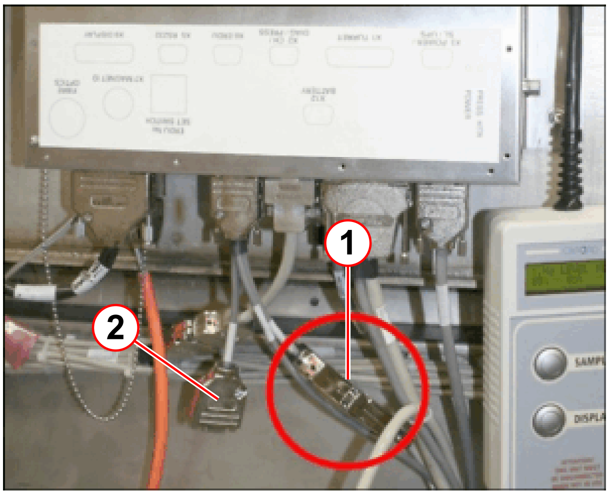
1. Disconnect the cable from the **flying connector** at the MSUP X6 ERDU / FIELD connector (**Avanto (early systems)** (→ Fig. 98 Page 99); **Espree/Avanto** (→ Fig. 99 Page 99)).

Fig. 98: Flying connector(s) on MSUP X6



(1) Flying connector on MSUP X6
(2) 2nd flying connector (optional 3rd Magnet stop button)

Fig. 99: Type B Flying connector(s) on MSUP X6



(1) Flying connector on MSUP X6
(2) 2nd flying connector (optional 3rd Magnet stop button)

» Check for Alarm Box buzzer and **MAG STOP** indicator **ON**.

» Check for MSUP Display indication **(10) ERDU BUTTONS "TOO FEW BUTTONS".**

» Reconnect the button cable to the flying connector and press the **ACKNOWLEDGE** button on the Alarm Box to silence the alarm.

» Check the MSUP display, window number 10, ensure it states "**ERDU BUTTONS OK**".

2. If the optional **(3rd) MAGNET STOP** button is installed, repeat the steps above for the **2nd flying connector**.

5.6.1.5 ERDU Operation Test

WARNING

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

» **DO NOT operate the Magnet Stop button on the ERDU Test Load unless an emergency magnet quench is required.**

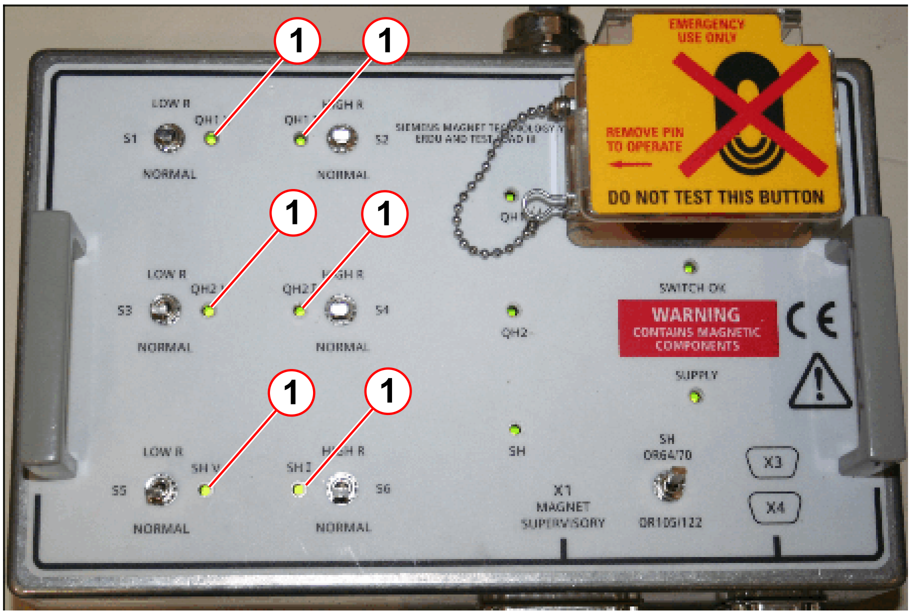
Thermal switches protect the load resistors in the ERDU Test Load from overheating. They will cut out after approximately three minutes of continuous operation. Keep the operating periods in the following tests as short as possible.

Table shows the various indications expected when the Magnet Stop or a remote button is pressed. The subsequent text describes the procedure in detail. Resistance values are nominal - +/-20% tolerance is acceptable.

Tab. 9 Magnet Stop or remote button is pressed

ERDU Test Load	MSUP-Display Unit	Alarm Box
QH1 V LED ON	7. SWITCH HEATER **260R A OFF A**	MAG STOP LED ON
QH2 V LED ON	8. QUENHTR **52R A 52R**	Buzzer ON
SH V LED ON	10. ERDU BUTTONS **ERDU OPERATED**	
QH1 I LED ON		
QH2 I LED ON		
SH I LED ON		

Fig. 100: ERDU Test load with ERDU pressed



(1) LEDs lit during ERDU activation

Refer to:(➔ Fig. 97 Page 97)

- Press the **ACKNOWLEDGE** button on the **Alarm box** to clear any indicators/alarms except on **ETL**.
- Scroll the **MSUP** display to #7.
- Press the **MAGNET STOP** button on the **Alarm Box**.
  - » Check that displays, indicators, and buzzers operate as in (➔ Tab. 9 Page 100).
- Release the **MAGNET STOP** button at the **Alarm Box**.
- Check the **MSUP Display** # 10 “ERDU BUTTONS OK”.
- Press **ACKNOWLEDGE** button on **Alarm Box** to clear any indicators/alarms except on **ERDU Test Load**.
- Scroll the **MSUP** display to #7.

- Press the remote **MAGNET STOP** button.
  » Check that displays, indicators, and buzzers operate as shown in (➔ Tab. 9 Page 100).
- Release the remote **MAGNET STOP** button.
- Check the **MSUP Display** # 10 “ERDU BUTTONS OK”.
- Press the **ACKNOWLEDGE** button on the **Alarm Box** to clear any indicators/alarms except on **ETL**.
- If installed, press the remote **optional (3rd) MAGNET STOP** button.
  » Check that displays, indicators, and buzzers operate as shown in (➔ Tab. 9 Page 100).
- Release the remote **MAGNET STOP** button.
- Check the **MSUP Display** # 10 “ERDU BUTTONS OK”.
- Press the **ACKNOWLEDGE** button on the **Alarm Box** to clear any indicators/alarms except on **ETL**.

5.6.1.6 Restore to normal operation

> Ensure all MAGNET STOP buttons are released and are NOT ACTIVE!

- Disconnect **X1** from the MSUP
- Check that the MSUP Display **(10) ERDU BUTTONS** is displaying the status **OK**.
- Disconnect the **X1** turret cable connector from **adaptor cable B** connection
- Reconnect **X1** turret cable connector to the **X1** connection on the MSUP
- Press the **ACKNOWLEDGE** button on the Alarm Box. Check that all indicators are now normal.
- The ERDU test generates some error / warning messages on the MRC. These error messages can be ignored.
- Click **< OK >** to confirm the messages.
- Click > System > Control > Tab MR Scanner > Reboot to reboot the scanner.

5.6.1.7 Final steps

- Disconnect the **PSU** from the AC supply.
- Disconnect the **ERDU Test Load** from its PSU.
- Re-pack all items in the **ERDU Test Load Kit box** to prevent loss.
- Disconnect the **MSUP-Display Unit**.

> The helium remote display unit is a Service Tool and must be disconnected during imaging to avoid interference to the MR Signal.

- Refit the **covers**.

- Affix the **tamper-proof seal** (Part no. **8100831**) to the **Alarm Box** .

Fig. 101: Tamper-proof seal location

(1) Tamper-proof seal

5.6.2 ERDU (MSUP K2306)

Required time: 15 min

Required tools: Screwdriver

SI Magnet Stop button test (ERDU)

5.6.2.1 Prerequisites

⚠ WARNING

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorised personnel must be excluded from the imaging suite.

» **Secure the room and prevent unauthorised access.**

- **Restrict access** to the magnet if it is at field when the test is being performed.
- **Check** that all alarm and error indications show that the **system is OK**.
- Remove the appropriate covers to gain access to the **MSUP** and **Service Magnet Stop button**.

Testing the Magnet Stop button (ERDU)

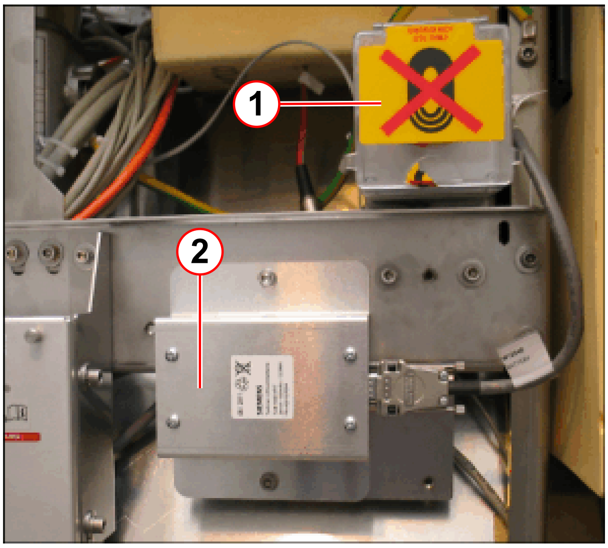
⚠ CAUTION

Magnet quench!

A magnet quench is caused by pressing the service magnet stop button (ERDU).

» **Do not operate the Service Magnet Stop button unless an emergency magnet quench is required.**

Fig. 102: Service Magnet Stop button



(1) Service Magnet Stop button
(2) ERDU battery

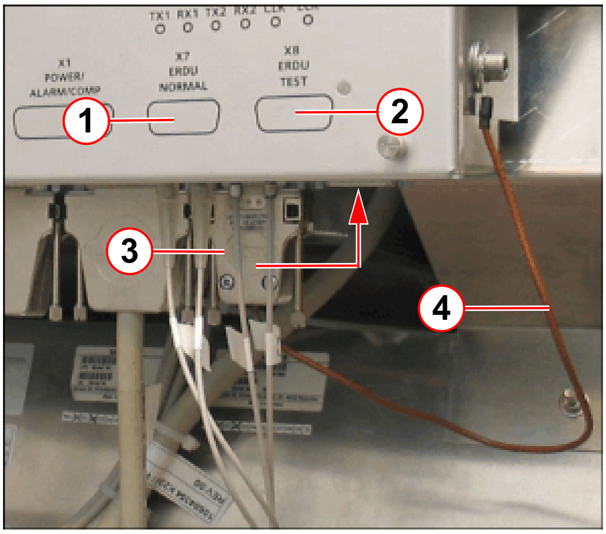
- The **Service Magnet Stop button** is mounted above the MSUP battery.
- Check that the button **is not activated.**
- **Test the Service Magnet Stop button**.
  » Press the **button** to ensure the **MSUP audible alarm** sounds.
  » **Reset** the button - ensure the alarm **stops**.

ℹ When the ERDU test connector is disconnected from BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET.

When the ERDU test connector is disconnected from the MSUP (X7), the customer site Magnet Stop button WILL NOT Quench the magnet. The MSUP audible warning is activated.

When ERDU test connector is connected to (X8), the MSUP audible warning continues. The Service Magnet Stop button is now active.

Fig. 103: ERDU test connector



(1) ERDU NORMAL
(2) ERDU TEST
(3) ERDU test connector
(4) ERDU connector strap

The **ERDU test connector** is fitted to **X7** during normal operation.

- Disconnect the test connector from **X7** (ERDU NORMAL).
  - » MSUP **audible alarm** is activated.
- Connect it to **X8** (ERDU TEST).
  - » On the **alarm box**, check the **audible alarm** sounds.
  - » Press **AUDIO ALARM OFF** to silence the alarm.
- Check that the **ERDU TEST LEDs** are **NOT illuminated**.

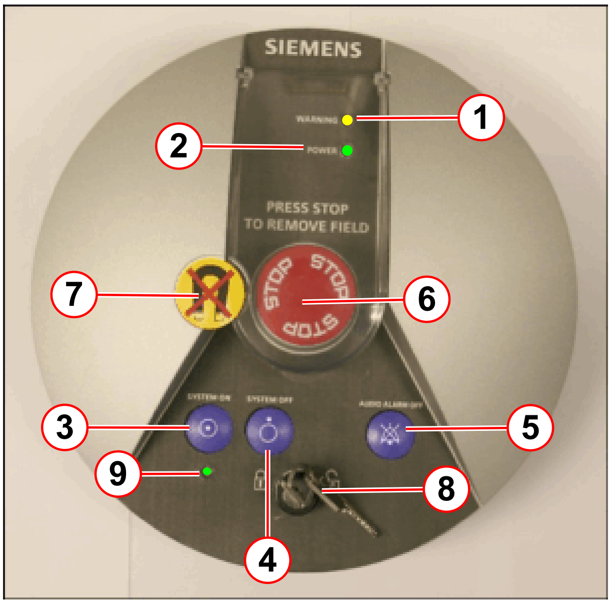
If the MSUP audible warning is not sounding, DO NOT CONTINUE.

Investigate the cause of the problem.

The Magnet Stop button locks when depressed.

Turn the button clockwise to unlock.

Fig. 104: Alarm box - LEDs and buttons

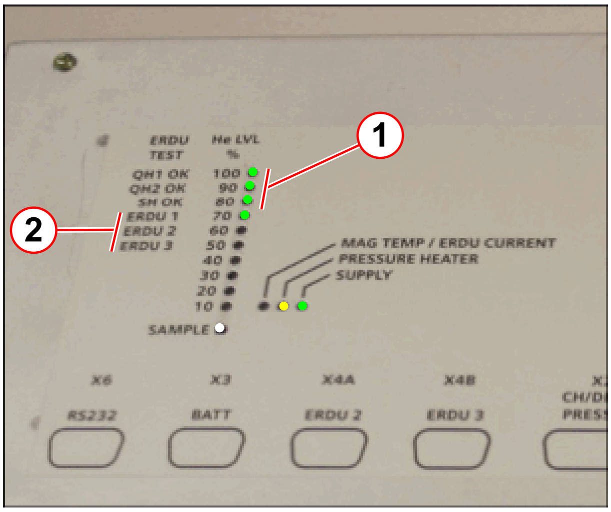


(1) LED WARNING = yellow
(2) LED POWER = green
(3) Button - SYSTEM ON
(4) Button - SYSTEM OFF
(5) Button - AUDIO ALARM OFF
(6) Button - MAGNET STOP (ERDU)
(7) Tamper-proof seal
(8) Power switch locking key
(9) SYSTEM ON LED = green

At the alarm box:

- Remove the tamper-proof seal (if required).
- Press the Magnet Stop button.
  » On the alarm box, check the audible alarm sounds and the yellow warning LED flashes.
  » Press AUDIO ALARM OFF to silence the alarm.

Fig. 105: ERDU test - LED indicators (typical)



(1) QH1, QH2, SH = green LED (all ON)
(2) ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

- Check QH1, QH2, SH LEDs are illuminated.
- Check ERDU 1 LED is illuminated.
- Reset the Magnet Stop button on the alarm box.
- Check the MSUP LEDs return to normal.
  » QH1, QH2, SH and ERDU 1 are NOT illuminated.

In the **Magnet Room**:

- Press the **Magnet Stop button**.
  - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.

  - Press **AUDIO ALARM OFF** to silence the alarm.
- At the **MSUP**:
  - Check **QH1, QH2, SH LEDs** are illuminated.
  - Check **ERDU 2 LED** is illuminated.
- **Reset** the **Magnet Stop button** in the Magnet Room.
- **Check** the LEDs return to **normal**.

An optional **3rd Magnet Stop button** may be fitted. Follow the steps below**:**

- Press the **Magnet Stop button**.
  - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.
  - Press **AUDIO ALARM OFF** to silence the alarm.
- At the **MSUP**:
  - Check **QH1, QH2, SH LEDs** are illuminated.
  - Check **ERDU 3 LED** is illuminated.
- **Reset** the **3rd Magnet Stop button**.
- **Check** the LEDs return to **normal**.

Final steps

- Check **ALL Magnet Stop buttons** are **not activated**.
- **Disconnect** the ERDU test connector **from X8** (ERDU TEST) and reconnect it to **X7** (ERDU NORMAL).
  - » On the **alarm box**, check the **audible alarm** sounds.
- **At the MSUP:**
  - » Check the **MSUP audible alarm** is **deactivated.**
  - » Check the **MSUP LEDs** return to **normal**.
- **At the alarm box:**
  - » Press **AUDIO ALARM OFF** to silence the alarm.
- **Check** the **warning LED** is not illuminated **'ON'**.

> i If the LED is on, or flashing, the fault must be checked.
> Do not disconnect the ERDU battery.

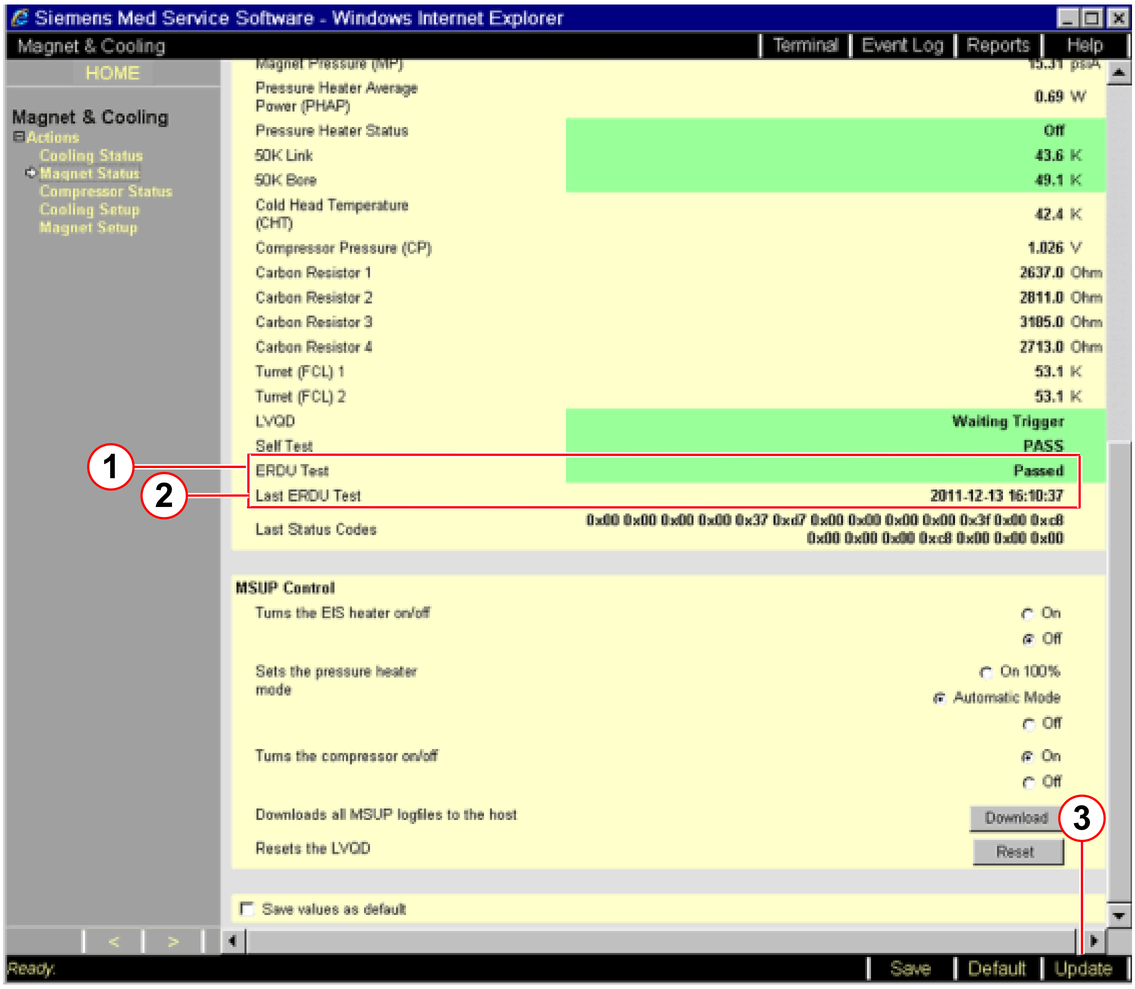
ERDU Test status

- Access the **Service Software** at the **MRC.**
- Select the menu item **Magnet & Cooling - Magnet Status**.

Data is only accurate from the last refresh of the page, or from when the page was opened last.

- Check the status of the **ERDU Test**(→ 1/Fig. 106 Page 107) and the date of the **Last ERDU Test**. (→ 2/Fig. 106 Page 107)

Fig. 106: ERDU Test status



(1) ERDU Test
(2) Last ERDU Test
(3) Update

5.7 RF Integrity

5.7.1 SAR Monitor Test

With SW Version VB15A and higher, the SAR Monitor Test is automated and part of 'General Quality Assurance'. There is no need to perform this test manually.

The manual test below ((→ Preliminary steps / Page 108) - (→ SAR Check / Page 110)) must be performed only for systems with SW Version VB13 or lower.

The purpose of this test is to verify that the SAR check is activated.

Required time: 10 min.

Required tools: Loaded body phantom, large spherical phantom, phantom holder 16 (P/N 7579613) for Espree only.

5.7.1.1 Preliminary steps

Preliminary steps for the measurements:

1. Press the **Home Position** key on the control panel to move the patient table to the “Home Position” (outermost position).
2. Press the **down/outward** key on the control panel to reset the display of the horizontal table position.
3. For Espree only; place phantom holder 16 on the table
4. Place the “loaded body phantom” and the “large spherical phantom” of the body coil on the table.
5. Press the **in/upward** key to move the patient table in until the display indicates:
   - for Avanto 300 mm ± 10 mm
   - for Espree 100 mm ± 10 mm
6. Activate the light localizer and shift the “loaded body phantom” with the “large spherical phantom” centered underneath.
7. Switch off the light localizer.
8. Press the **center position** key to move the table to the center position.

5.7.1.2 Patient registration:

Register patients by supplying the following data:

Last name:	**SarCheck** ↵
Patient ID:	**1** ↵
Date of Birth:	Today’s date (enter the date according to the regional date format setting, e.g., **mm-dd-yy** ↵)
Sex:	**Male**
Age:	

Weight (lb): **200 lb** ↵

or

Weight (kg) **90.8 kg** ↵

Study: **select any protocol**

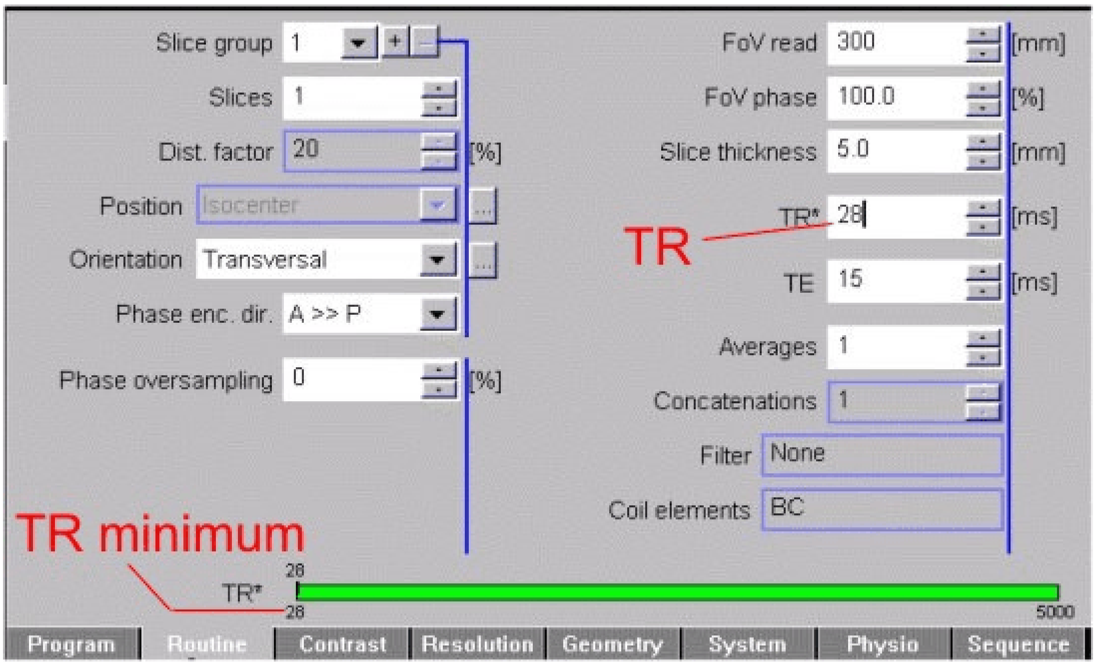
Patient position: **Head first - supine**

- Click **< Exam>**
  - » The **"Confirmation"** dialog box is displayed.
- Click **<Confirm>** to confirm the patient data.
  - » The **"Warning"** window ......`Do you want to change the patient data?` opens.
- Click **< No>** to register the patient data.

5.7.1.3 Protocol setting

- Select the **Exam** task card.
- Select **View > Exam Explorer** in the main menu
  - » The Exam Explorer opens
- Select sequence se_15b130 (**Siemens > Sequence Region > Siemens Sequences > Default Protocols > se_15b130**; and drag and drop to the bottom left).
- Click **< Open >** and input the following protocol settings
  - » TR = **TR** $_{\text{minimum}}$

Fig. 107: TR minimum



5.7.1.4 SAR Check

SI Checking the SAR monitor (only for systems with SW version VB13 or earlier)

Click **< Apply >** to start the sequence.

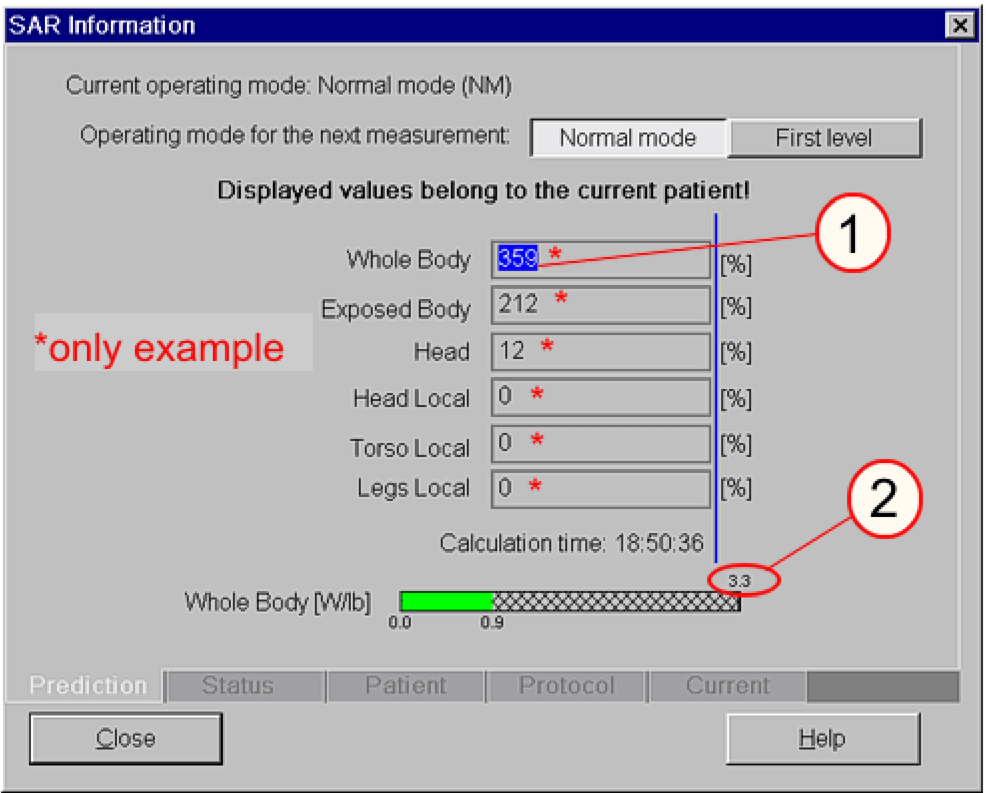
» If the pop-up window “Automatic Table Move” opens, press **< Continue >** to close the window.

» If the pop-up window “`SARLimit(s) Exceeded!`” opens, press **< Cancel >** to close the window. (Whether the pop-up windows appears is irrelevant to the successful performance of this test.)

- Select **Option > SAR Information** in the main menu.

» The “SAR Information” window opens on the SAR Info tab card.

Fig. 108: Example of “SAR Information” window



- Click the “**Whole body**” field with the left computer mouse button.

» The current Whole Body value is displayed **above** the progress bar; refer to 2 (→ Fig. 108 Page 110). The Whole Body value has to be greater than **0.2 W/kg** or **0.45 W/lb** .

WARNING

If the test for the SAR monitor is not completed successfully, ensure that all patient data is entered correctly.

» **If data is correct but the test is not completed successfully, reinstall the syngo MR software. Repeat the SAR monitor test.**

5.7.2 Measurement of body Coil Power Loss

Required time: 5 min

Required tools: Small spherical phantom with the holder

Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Preliminary steps:

1. Position the small spherical phantom in the magnet isocenter (without loader).
2. Start the Service platform **Options > Service > Local Service**.
3. Select the Tune-up platform.

SI Measurement of the Coil Power Loss

1. Select the **Coil Power Losses** measurement only.
2. Start the **Coil Power Losses** measurement by clicking the **Go** button.

   » The measurement has been completed successfully, if the **Coil Power Losses** Tune-up step indicates the status **'done'**.

5.8 Quality Assurance (QA)

Required time: 70 - 120 min depending on installed options.

Required tools: QA Phantom Setup, Service Plug (only Avanto and, if Multi Nucleus Spectroscopy Option is installed)

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.

> Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the ‘Help’ button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Start the Service platform **> Options > Service > Local Service**.

SI General Quality Assurance

Go to the Home menu and press the corresponding button to access the **Quality Assurance** sub platforms.

- Start the measurement by clicking the “Go” button.
  - » The **General Quality Assurance** check is successfully completed when every individual QA step indicates the status **< done >**.

> If the results are out of tolerance, perform “Tune-Up” or troubleshooting, if necessary. Repeat “Quality Assurance”.

5.9 End of Function Tests

End of section.

Initial Version

These instruction and the (→ Preventive Maintenance / M6-020.831.05) replace the following document:

- Maintenance Instructions Combined Workflow
- The maintenance instruction (→ Preventive Maintenance / M6-020.831.05)
- Safety- related test (these document) (→ M6-020.833.05 / Safety-Related Tests according to IEC 62353 / Page 1)

Chapter	Changes
All	The procedures in these instructions are taken over from the document 'Maintenance Instructions Combined Workflow' with minor changes. ▪ Editorial changes

Version 02 compared to Version 01

Chapter	Changes
K02	Identification of the controlled access area: (→ Identification of the controlled access area (0.5 mT area) / Page 35) ▪ Editorial changes

There are no Hazard IDs in this document.

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